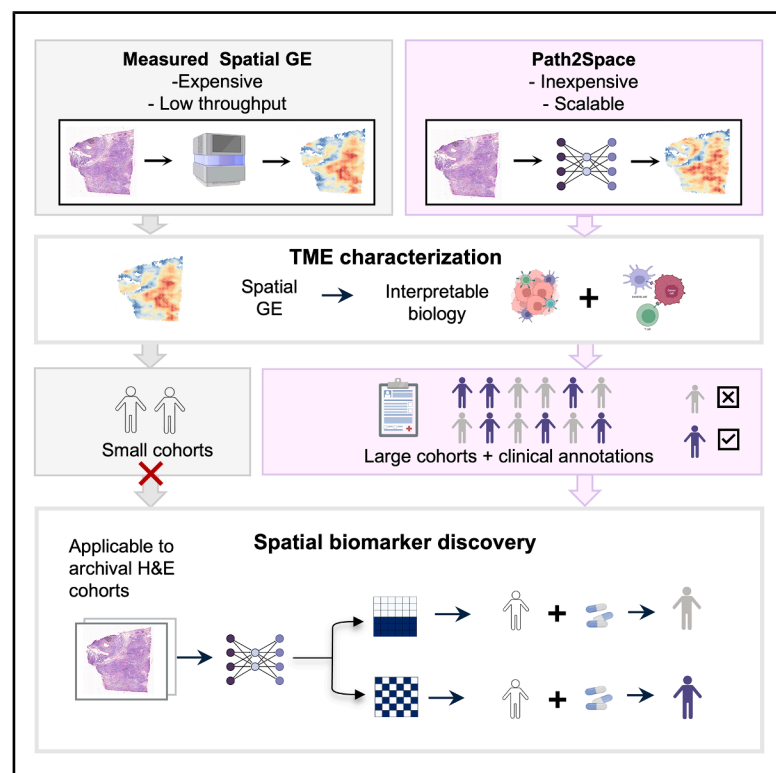


AI-predicted spatial transcriptomics unlocks breast cancer biomarkers from pathology

Graphical abstract



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In brief

Path2Space predicts spatial gene expression directly from histopathology, enabling low-cost, large-scale characterization of the tumor microenvironment. The inferred spatial breast cancer landscapes reveal clinically relevant subgroups and improve the prediction of treatment response from routine H&E slides alone.

Highlights

- Path2Space predicts the spatial expression for thousands of genes from H&E slides
- Spatial patterns define breast cancer subgroups with distinct biology and survival
- Spatial biomarkers outperform bulk sequencing for therapy response
- Path2Space enables scalable discovery of response biomarkers from routine histopathology

Article

AI-predicted spatial transcriptomics unlocks breast cancer biomarkers from pathology

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SUMMARY

Spatial transcriptomics (ST) assays are transforming our understanding of tumor heterogeneity, but their high cost limits their application in large-scale biomarker discovery. Here, we present “Path2Space,” a deep-learning model that predicts spatial gene expression directly from histopathology slides. Trained on extensive breast cancer ST data, Path2Space robustly predicts the spatial expression of thousands of genes, outperforming 21 established methods. Charting the tumor microenvironment (TME) of 976 breast cancer TCGA (The Cancer Genome Atlas) tumors, it accurately infers cell-type abundances and identifies three spatially defined breast cancer subgroups with distinct survival outcomes. Notably, the derived low-cost spatial TME landscapes enable more accurate predictions of patient response to chemotherapy and trastuzumab compared with costly conventional bulk-sequencing-based biomarkers. Path2Space thus offers a scalable, fast, and cost-effective alternative to molecular assays. It opens avenues for large cohort treatment biomarker discovery and translationally relevant insights into tumor biology, with potential applicability across many cancer indications.

INTRODUCTION

Spatial transcriptomics (ST) is transforming our understanding of breast cancer by enabling the mapping of gene expression within intact tumor tissues. It reveals intratumoral heterogeneity and the architecture of the tumor microenvironment (TME). Recent ST studies characterize spatially distinct tumor cell populations and stromal-immune niches, including macrophage subsets linked to clinical outcomes.¹ ST also identifies chemokine-rich blood vessel niches promoting lymphocyte infiltration, with higher vessel density correlating with improved clinical outcomes.² In HER2-positive breast cancers, ST delineates spatial interactions between T cells and macrophages, highlighting a type I interferon response associated with these immune niches.³ In metastatic lesions, ST distinguishes immune-excluded from T cell-infiltrated regions and epithelial-mesenchymal transition phenotypes, providing insights into metastasis and therapy resistance.⁴

Yet, the translational potential of ST remains constrained by its high cost.⁵ This limits large cohorts needed for biomarker discovery and validation. Because ST data are often paired with hematoxylin and eosin (H&E) images, predicting spatial gene expression from H&E slides offers a scalable alternative. Deep learning applied to H&E whole-slide images (WSIs) identifies cell types⁶ and molecular features, including genetic mutations,⁷ bulk mRNA expression,⁸ and methylation patterns.⁹ Previous efforts to predict spatial gene expression from H&E images are limited to a small subset of genes,^{10–18} precluding their application to survival and treatment response analyses in large clinical cohorts.

To address this challenge, we developed “Path2Space,” a model that predicts spatial gene expression from histopathology slides, focusing on breast cancer due to its prevalence and rich data availability. The model demonstrates robust predictive performance on external datasets and can reliably infer localized cell-type compositions from the associated predicted spatial gene expression patterns. When applied to the breast cancer cohort from The Cancer Genome Atlas (TCGA), Path2Space identifies three prognostic breast cancer subgroups based on their predicted spatial transcriptomic cluster compositions, a classification framework we term SpatioTypes. Applied to four cohorts of breast cancer patients treated with trastuzumab and three treated with chemotherapy, Path2Space uncovers spatial biomarkers and predicts therapy response at accuracy levels higher than or equivalent to those obtained from measured bulk tumor transcriptomics, without any need for costly and labor-intensive tumor sequencing.

RESULTS

Study overview

We developed Path2Space, a deep-learning model to predict spatial gene expression from H&E WSIs of breast cancer tumors to identify microenvironmental features and spatial biomarkers associated with treatment response.

To develop the spatial gene expression predictor, we trained a deep-learning model on 10× Genomics Visium ST data from slides with matched H&E images, in which gene expression profiles are measured from 55-μm-diameter capture areas

(“spots”). We predicted 14,068 genes detected in ≥5% of spots. The prediction model consists of two main components: preprocessing and regression (Figure 1A). During preprocessing, tile images surrounding individual spots are extracted, color-normalized, and processed using CTransPath, a foundational digital pathology model for feature extraction¹⁹ (Figure S1A; STAR Methods). In the regression step, a multilayer perceptron (MLP) neural network is trained to predict the gene expression vector (14,068 genes) for each spot from its corresponding tile image. Prediction targets were log-transformed counts, which outperformed counts per million (CPM)-based models, yielding improved gene prediction and cell-type-specific signal capture (Figures S1B and S1C; STAR Methods).

The model was trained using a leave-one-patient-out cross-validation framework with ensemble prediction on the Bassiouni et al. cohort,²⁰ comprising 56,567 matched image-expression spot pairs (Figures S1D and S1E; STAR Methods).

Following model training, we applied a spatial smoothing technique to mitigate technical variability in ST data (Figures S1F–S1H; STAR Methods). This approach averages the predicted gene expression values of each spot with its immediate neighbors. Furthermore, aiming to mitigate measurement noise,^{21–24} we applied the same smoothing procedure to the measured expression values. Both smoothed predictions and smoothed measurements were used for evaluating prediction performance (Figure 2), and smoothed predictions were used for all downstream analyses. For benchmarking against other methods (Figure 3), smoothed predictions were evaluated against unsmoothed measured expression to ensure fair comparison across methods. Results using unsmoothed predictions are further reported in Figures S3A and S4 and Table S2.

Predictive accuracy varied across slides and correlated strongly with image quality metrics, including sharpness and H&E staining intensity. These relationships informed quality control (QC) thresholds for image and transcriptomic data, including total, mitochondrial, and hemoglobin read counts per spot (Figure S2; STAR Methods). After applying these QC thresholds, 22 tissue sections from 14 patients were retained.

The model was subsequently evaluated on three independent external ST validation cohorts. After QC, test sets comprised five samples from four patients in HEST,²⁵ four samples from four patients in Martinez et al.,²⁶ and nine samples from seven patients in the Human Tumor Atlas Network (HTAN).²⁷

Finally, the primary objective of Path2Space was to apply the trained ST model to discover *spatial biomarkers* of patient survival and treatment response from pathology slides in large patient cohorts. We applied Path2Space to the TCGA breast cancer cohort (1,096 samples from 976 patients, of whom 853 had complete survival data), identifying and characterizing spatial architectures associated with disease-free survival (Figure 1B). We further applied Path2Space to study TMEs and infer treatment response biomarkers for chemotherapy and trastuzumab in four additional breast cancer cohorts with treatment response data: TransNEO²⁸ (chemotherapy $n = 93$, trastuzumab $n = 61$), the Personalised Breast Cancer Programme (PBCP)²⁸ (chemotherapy $n = 19$, trastuzumab $n = 18$), IMPRESS²⁹ (chemotherapy $n = 64$, trastuzumab $n = 62$), and Cedars-Sinai (trastuzumab $n = 26$) (Figure 1C; Table S1).

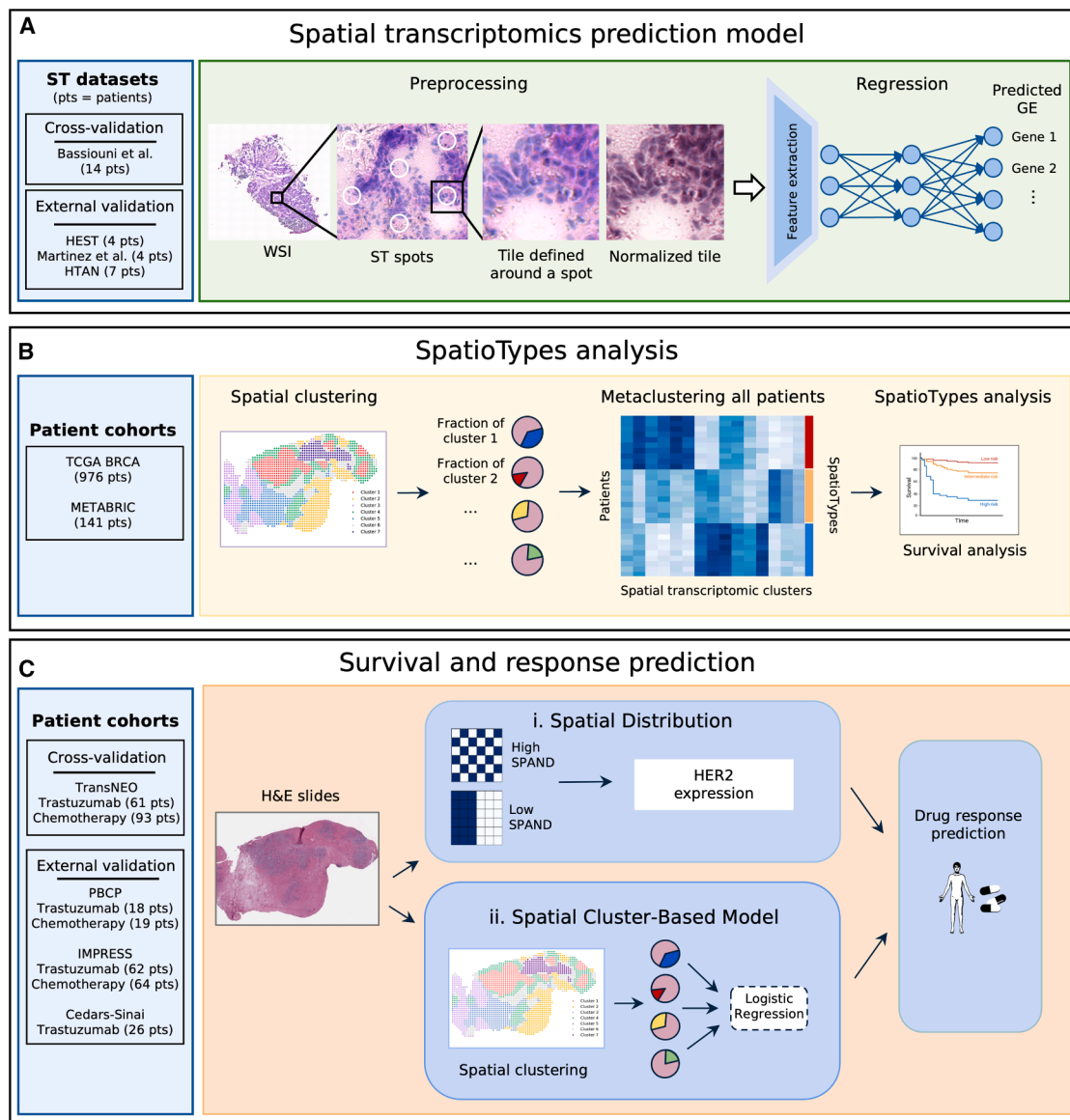


Figure 1. Overview of datasets and computational workflow

(A) Path2Space was trained and evaluated on four paired H&E-ST datasets. WSIs were tiled around ST spots, color-normalized, encoded using the CTransPath foundation model, and passed to a multilayer perceptron (MLP) to infer spot-level gene expression.

(B) Path2Space was applied to TCGA breast cancer (BRCA) WSIs to predict spatial gene expression and identify spatial clusters defining tumor spatial profiles, which were grouped using an unsupervised approach into prognostic breast cancer subgroups termed SpatioTypes.

(C) Outcome prediction via (i) spatial neighborhood diverseness (SPAND), quantifying HER2 spatial expression heterogeneity, and (ii) spatial cluster-based logistic regression, trained in cross-validation on TransNEO and validated on IMPRESS, PBCP, and Cedars-Sinai cohorts.

Predicting spatial gene expression from H&E slides

We trained and evaluated a Path2Space breast cancer model using the Bassiouni et al. Visium ST cohort, consisting of H&E images with matched gene expression profiles. Gene-wise accuracy was quantified using the Pearson correlation coefficient (PCC) between predicted and measured expression values across all spots, evaluated both with and without spatial smoothing applied to predictions. For each gene, we calculated PCC values for each slide, then averaged these values within each held-out patient, and finally across all patients, yielding a median gene-wise correlation

of 0.38 (0.20 unsmoothed; Figure 2A). Of 14,068 genes, 6,629 exceeded a mean PCC > 0.4 with smoothing and 1,376 without (Figure S3A). Binary classification of high versus low expression (relative to each slide's median) produced median AUC values of 0.70 (smoothed) and 0.65 (unsmoothed), with 3,116 genes (smoothed) and 415 (unsmoothed) surpassing 0.75. Additional metrics confirmed robust performance (Table S2).

To assess the generalizability of the model's predictive power, we applied the Bassiouni et al.-trained model to three independent ST cohorts (HEST, Martinez et al., and HTAN), using QC

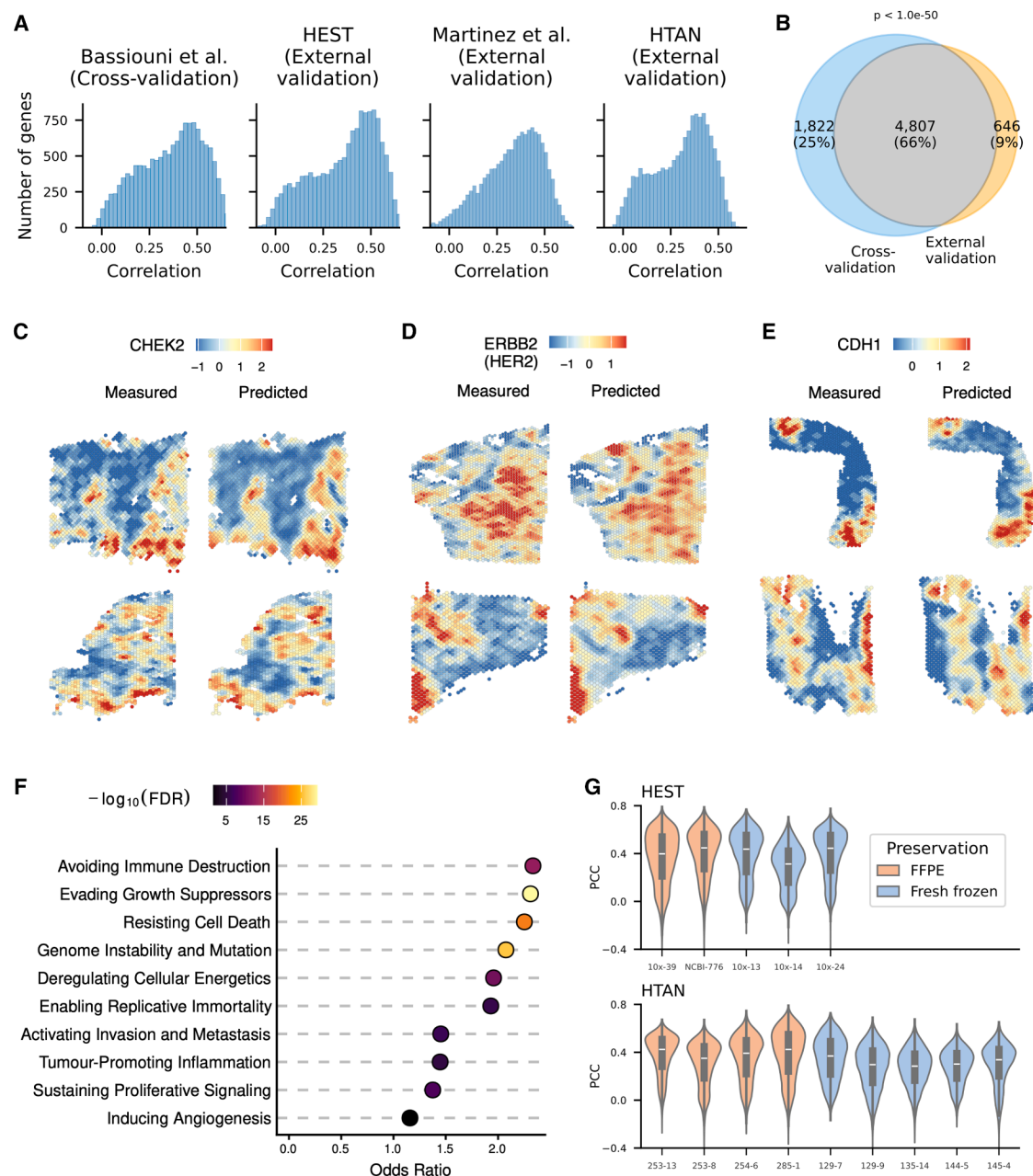


Figure 2. Performance of Path2Space spatial gene expression prediction

(A) Distribution of gene-wise PCCs between predicted and measured expression in Bassiouni et al. (cross-validation) and HEST, Martinez et al., and HTAN (external validation) cohorts.

(B) Overlap between genes with PCC > 0.4 in cross-validation and those with PCC > 0.4 averaged across validation cohorts. p value by hypergeometric test.

(C–E) Representative robustly predicted, clinically relevant genes: *CHEK2* (sections 395A and 094D), *ERBB2/HER2* (093C and 395C), and *CDH1* (398A and 095B). For each gene, measured (left) and predicted (right) expression are shown for two Bassiouni sections, with spots colored by scaled expression.

(F) Gene set enrichment of the 10 cancer hallmark sets among robustly predicted genes (defined in B). Color indicates $-\log_{10}$ adjusted p values (all FDR ≤ 0.05).

(G) Gene-wise PCC distributions in HEST and HTAN, stratified by tissue preservation (FF vs. FFPE).

See also [Figures S1–S3](#)

thresholds fixed a priori based on the training cohort. Using the same hierarchical gene-wise PCC framework, median PCC and AUC values were as follows: HEST, 0.40 (0.22 unsmoothed) and 0.71 (0.64), with 7,023 (1,268) genes > 0.4; Martinez et al.,

0.36 (0.20) and 0.69 (0.66), with 5,409 (611) genes > 0.4; and HTAN, 0.33 (0.17) and 0.67 (0.62), with 4,401 (291) genes > 0.4 (Figure 2A). These results demonstrate robust generalization across cohorts. Performance saturated with

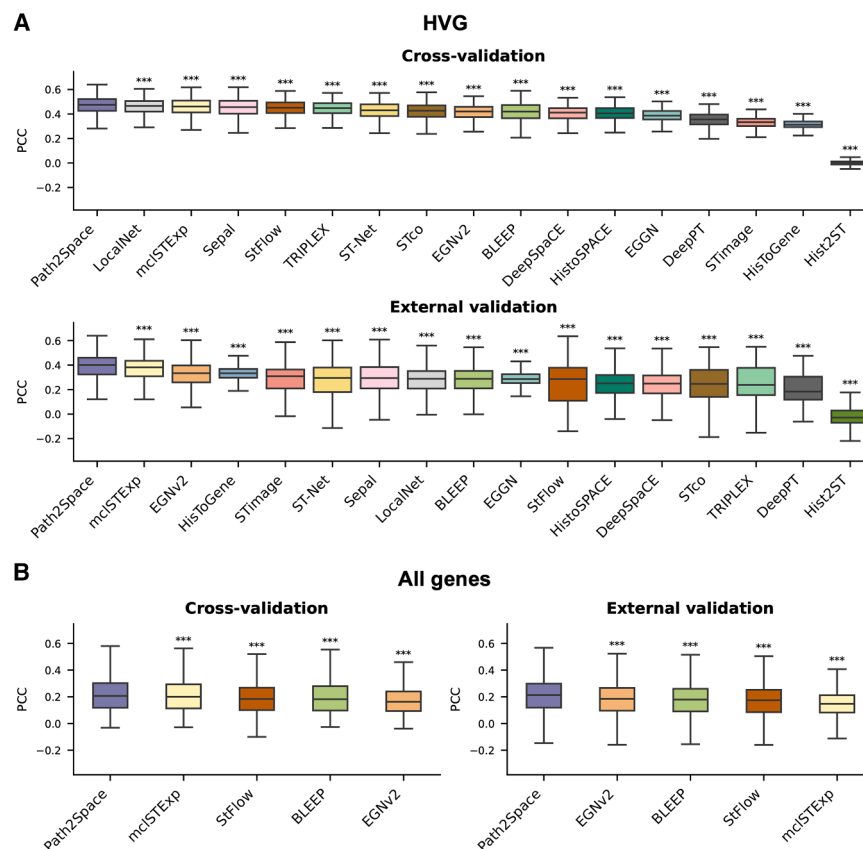


Figure 3. Benchmarking against state-of-the-art ST prediction models

(A) Comparison of Path2Space against 16 other ST prediction models trained on Bassiouni et al. All models were trained on unsmoothed expression data, and evaluation was performed against unsmoothed measured expression. Per-gene PCCs for 988 HVGs are shown for cross-validation (top) and external validation averaged across the HEST, Martinez, and HTAN cohorts (bottom). Models ordered by median PCC.

(B) Extension to all 14,068 genes for the top five models (Path2Space, EGNv2, mciSTExp, BLEEP, and StFlow) using the same evaluation setup. Statistical significance was assessed using a two-sided Mann-Whitney U-test comparing each method to Path2Space (**p < 0.01; ns, not significant). Error bars represent the interquartile range, with the central bar indicating the median. See also Figure S4

Bassiouni et al. cohort, we identified highly variable genes (HVGs, $n = 988$) following a previously published benchmarking protocol³⁰ (STAR Methods). All models were trained in cross-validation on this Visium breast cancer dataset, and performance was quantified using PCCs between predicted and raw, unsmoothed measured expression values.

Path2Space with spatial smoothing (comparing smoothed predictions to unsmoothed measured expression) achieved the highest median PCC in both cross-validation and external validation, significantly outperforming all other models ($p < 0.001$; Figure 3A; Table S3). Notably, while spatial smoothing significantly improved Path2Space's performance, even without this post-processing step, Path2Space remained among the top-performing methods (unsmoothed predictions vs. unsmoothed labels; Figure S4A). These results were robust to normalization strategy: when all models were trained on log-transformed CPM-normalized counts, Path2Space again achieved the highest median PCC (Figure S4B). To compare against published benchmarks (bringing the total number of unique benchmarked methods to 21), we evaluated Path2Space on a legacy (pre-Visium) ST dataset³ using the 785 HVGs and cross-validation protocol from the published benchmark.³⁰ Compared against eleven other models, Path2Space achieved the second-highest median PCC, with performance statistically indistinguishable from the top-performing method, EGNv2³¹ (Figures S4C and S4D). Extending beyond HVGs, retraining top models on all genes ($n = 14,068$) showed that Path2Space achieved the highest PCC ($p < 0.001$; Figures 3B and S4E).

H&E-inferred ST captures cell-type-specific expression and abundance

Having established prediction accuracy on ST-paired slides, we next assessed whether Path2Space captures spatial

modest training size and generalized across breast cancer subtypes (Figures S3B–S3F; STAR Methods).

Remarkably, we observed substantial concordance in highly predicted genes between cross-validation and external validation, with 4,807 genes achieving correlations >0.4 in both settings (Figure 2B). This group included genes that are crucial in breast cancer prognosis and treatment decisions, such as *CHEK2*, *ERBB2* (HER2), and *CDH1* (Figures 2C–2E). Gene set enrichment analysis (GSEA) of the 4,807 robustly predicted genes revealed significant enrichment (false discovery rate [FDR] < 0.05) for all 10 cancer hallmark processes. Notably, pathways related to avoiding immune destruction, evading growth suppressors, and resisting cell death showed the strongest enrichment (Figure 2F; Table S2). Gene predictability strongly correlated with detection rate (PCC = 0.70), and genes with unexpectedly low predictability were enriched for secretory and extracellular components (Figure S3G; Table S2).

Despite Path2Space being trained exclusively on fresh frozen (FF) slides, performance was similar across tissue preservation methods in external validation (median PCC: FF = 0.35, formalin-fixed paraffin-embedded (FFPE) = 0.37; linear mixed-effects model adjusting for cohort, $p = 0.69$; Figure 2G), indicating that Path2Space generalizes effectively to FFPE slides, enabling biomarker discovery in archival tissue repositories (STAR Methods).

To contextualize these results, we benchmarked Path2Space against 16 state-of-the-art ST prediction models. Using the

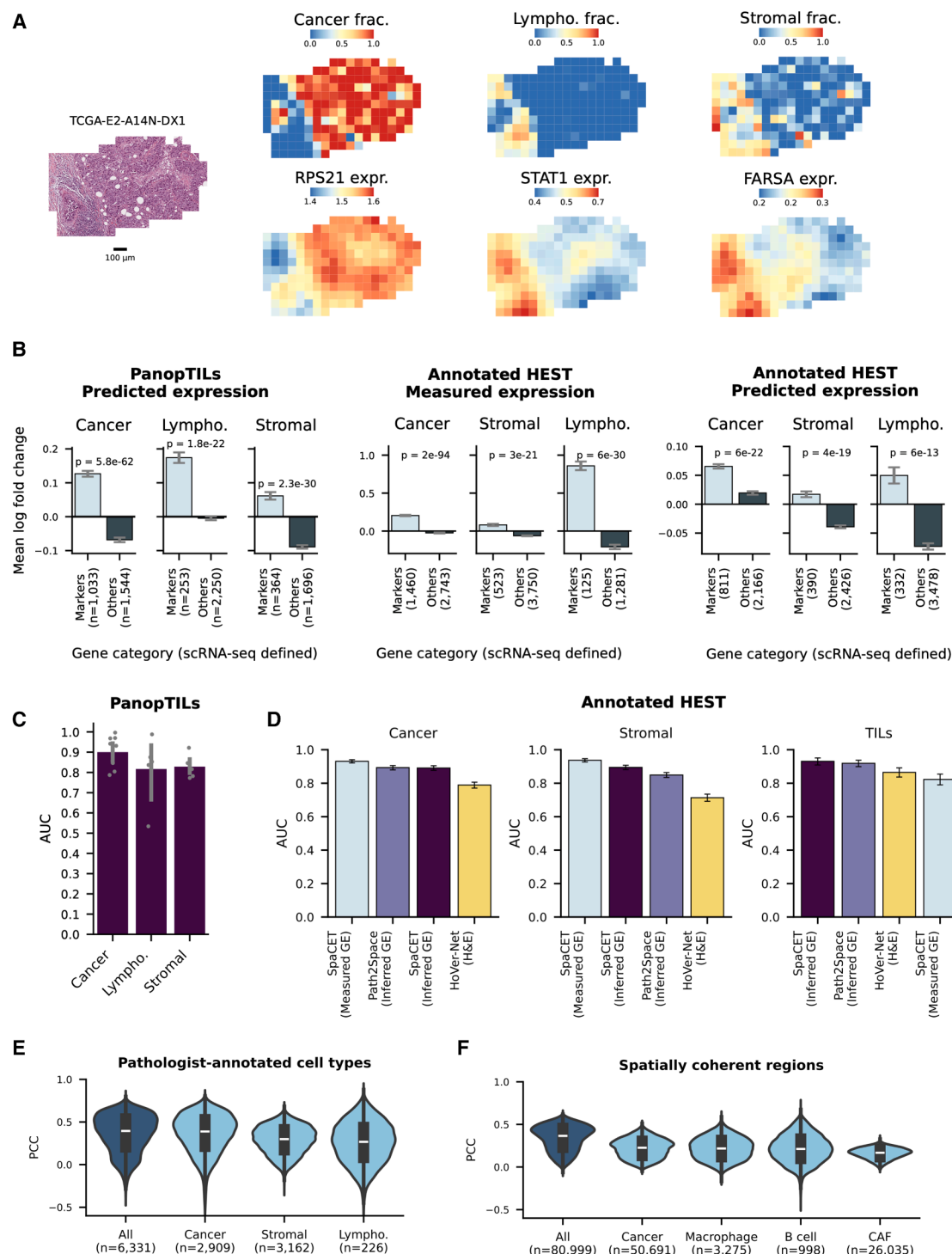


Figure 4. Evaluation of Path2Space predictions and cell-type inference performance

(A) Example Path2Space predictions for cell-type-specific marker genes (Bassez et al.) on a PanopTILs-annotated TCGA slide (TCGA-E2-A14N-DX1). Heatmaps show spot-level PanopTILs cell-type fractions (top) and scaled predicted expression of corresponding marker genes (bottom; scale bar, 100 μ m). (B) Differential expression (DE) of scRNA-seq-defined marker versus non-marker genes across major cell types, comparing enriched (>50%) versus absent (0%) regions. Left: Path2Space-predicted expression from PanopTILs-annotated TCGA slides. Center: measured ST from two pathologist-annotated HEST samples. Right: Path2Space-predicted expression for the same HEST samples. Bars show mean log fold change; two-sided Mann-Whitney U-tests. (C) Cell-type classification performance in the PanopTILs cohort using SpaCET on Path2Space-predicted ST. Bar plots show AUCs for cancer, lymphocytes, and stroma; points indicate per-slide AUCs ($n = 20$); and error bars show standard error.

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cell-type-specific expression patterns and enables accurate cell-type abundance estimation.

The PanopTILs³² dataset, 1,709 regions of interest (ROIs) from 151 patients, provides nucleus-level pathologist annotations for standard archival TCGA diagnostic slides—images without paired ST—representing the clinically relevant setting for biomarker discovery (Figure S5A). By contrast, the HEST dataset provides spot-level binary cell-type annotations on slides with paired Visium ST, enabling us to assess how closely Path2Space approaches the accuracy achievable with measured ST.

We validated Path2Space on PanopTILs through two complementary approaches. First, we inferred localized gene expression across annotated ROIs, focusing on cancer cells, lymphocytes, and stromal cells. Differential expression analysis comparing regions predominantly composed (>50%) of each cell type to regions lacking that cell type recapitulated patterns observed in single-cell RNA sequencing (scRNA-seq) data,³³ with marker genes showing significantly stronger upregulation in their corresponding enriched regions than non-marker genes (two-sided Mann-Whitney U-test, $p < 0.01$; Figures 4A and 4B). Second, we applied SpaCET³⁴—an unsupervised deconvolution method that estimates cell-type abundances from ST using scRNA-seq reference signatures—to Path2Space-inferred expression. When applied to H&E-inferred expression, SpaCET-derived fractions achieved PCCs of 0.69, 0.60, and 0.41 for cancer cells, lymphocytes, and stromal cells, respectively, and accurately classified regions as containing or lacking each cell type with AUCs of 0.85, 0.80, and 0.85 (Figures 4C and S5B).

To improve cell-fraction estimation, we trained a supervised MLP regressor on PanopTILs pathologist-annotated fractions. In 5-fold cross-validation, this supervised model consistently outperformed SpaCET across all cell types, achieving higher correlations (PCC = 0.80, 0.80, and 0.60) and classification accuracy (AUC = 0.88, 0.87, and 0.85) for cancer cells, lymphocytes, and stromal cells, respectively (Figure S5B). GSEA confirmed biologically meaningful predictions (FDR < 0.05; Figure S5C), indicating that the model learned biologically meaningful features.

To benchmark against measured ST, we evaluated Path2Space on two annotated HEST slides. Differential expression analysis confirmed that both measured and inferred expression showed significantly higher levels of cell-type-specific marker genes in their corresponding enriched regions compared with non-marker genes ($p < 0.01$ for all comparisons; Figure 4B). We then applied the PanopTILs-trained supervised model and SpaCET to Path2Space-inferred expression and compared performance to SpaCET applied to measured ST (Figure 4D). Path2Space-based approaches achieved AUCs of 0.85–0.89 for cancer and stromal cells, approaching SpaCET applied to

measured ST (AUCs of 0.93–0.94), and outperformed it for lymphocytes (AUCs of 0.92–0.93 vs. 0.82). Path2Space also outperformed HoVer-Net,³⁵ a direct H&E-based nuclear classifier, across all three cell types and achieved the highest deconvolution-based cell-type classification accuracy when compared with the top-performing ST prediction models from our benchmarking analysis (Figure S5D). More broadly, deconvolution performance based on Path2Space-inferred ST closely tracked that of measured ST across cell types, with both approaches showing abundance-dependent accuracy that declined for rarer populations (Figures S5E and S5F; Discussion).

Path2Space resolves gene expression variation within individual cell types

Having established an accurate prediction of cell-type-specific expression, we next evaluated whether its predictions resolve variation within individual cell types. Within pathologist-annotated cancer, stromal, and lymphocyte regions in the HEST validation cohort, median PCCs between predicted and measured expression ranged from 0.27 to 0.39 (cancer 0.39, stromal 0.30, and lymphocyte 0.27; Figure 4E), with 3,286–5,774 genes exceeding PCC > 0.4 per cell type, comparable to 5,894 genes in whole-tissue analysis. Similar results were observed in transcriptomically defined neighborhoods. Correlations within these neighborhoods showed median PCCs of 0.19–0.29 across validation cohorts, with over 2,000 genes exceeding PCC > 0.4 in four of five cell types (Figure 4F). Gene-wise predictive accuracy was highly concordant across cell types and both annotation approaches, with pairwise correlations of 0.52–0.97 (Figure S5G).

To test whether predictions capture variation beyond cell-type identity, we compared accuracy for the scRNA-seq-defined marker versus non-marker genes described above³³ (Figures 4A and 4B). Non-marker genes achieved slightly higher correlations than marker genes (median PCC = 0.39 vs. 0.36; $p = 2.6 \times 10^{-12}$, two-sided Mann-Whitney U-test; Figure S5H), demonstrating that prediction accuracy is not driven by canonical cell-type markers.

Path2Space defines three spatial breast cancer subgroups with distinct TMEs and survival

Having established accurate prediction of spatial gene expression and cell-type composition from H&E, we applied the model to the TCGA breast cancer cohort to characterize TME heterogeneity and identify spatial prognostic biomarkers. To match Visium resolution, we divided each TCGA slide into contiguous pseudo-spots (~6,131 per slide) and inferred spatial gene expression for each.

To identify common spatial transcriptomic patterns across TCGA, we applied SpaGCN,³⁶ a graph convolutional network,

(D) Comparison of deconvolution performance on the HEST samples. AUCs are shown for the Path2Space deconvolution model, SpaCET on measured ST, SpaCET on Path2Space predictions, and HoVer-Net. Error bars indicate 95% confidence intervals (1,000 bootstrap iterations).

(E and F) Gene-wise PCC distributions between predicted and measured expression across regions corresponding to each cell type. (E) Performance across pathologist-annotated HEST regions assigned to each cell type. (F) Performance across regions defined by unsupervised clustering of spatial proximity and transcriptomic similarity and labeled by dominant cell type across validation cohorts. Violin plots combine cross-validation and validation samples.

See also Figure S5

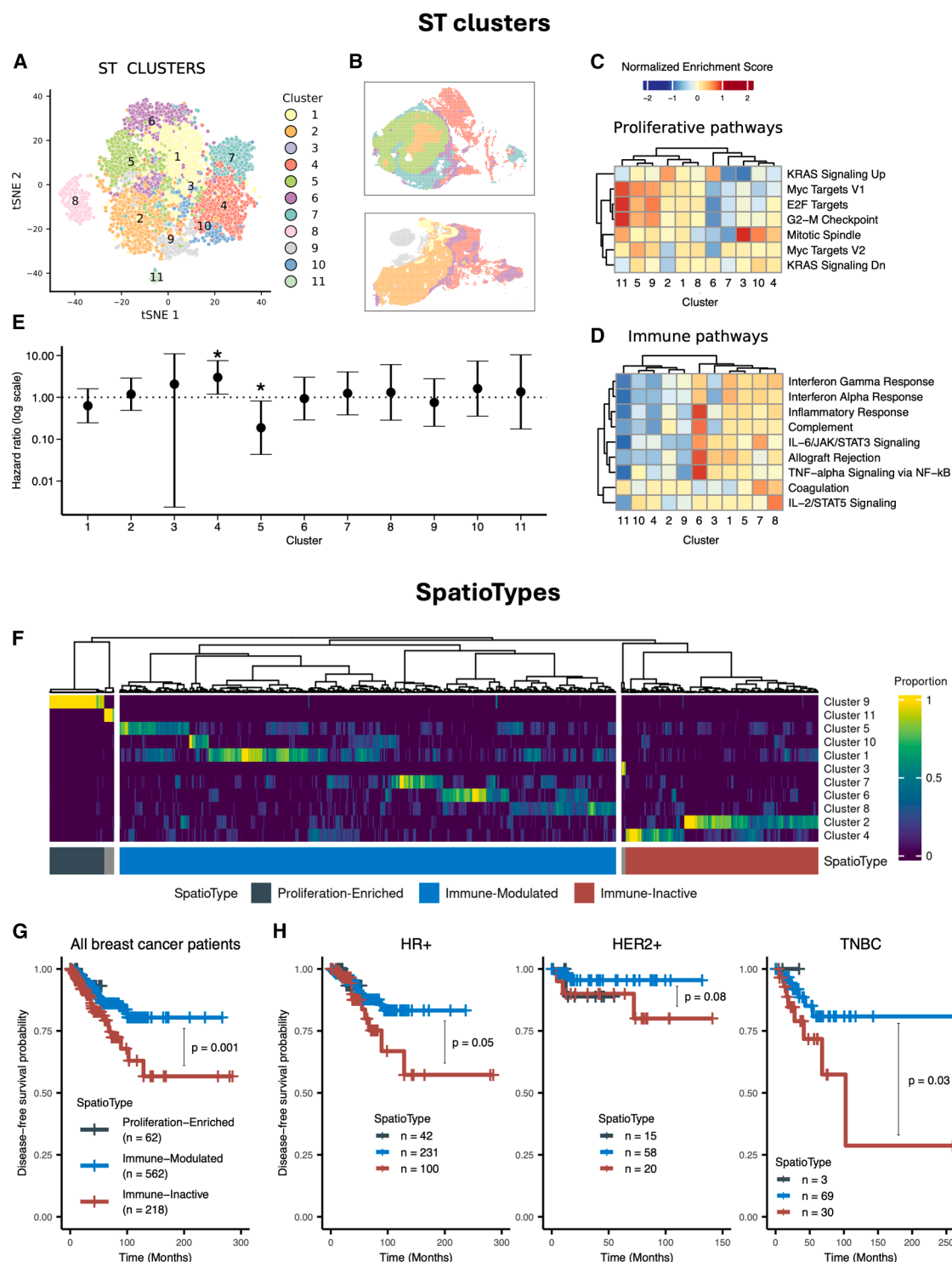


Figure 5. Identification of prognostic tumor SpatioTypes in TCGA-BRCA

(A) t-SNE of spatial domain-averaged gene expression across TCGA slides reveals 11 ST clusters. Each point represents a spatial domain identified by SpaGCN ($n = 7,295$ domains from 1,096 slides; average 6.6 per slide), colored by ST cluster.

(B) Representative TCGA slides showing spatial distributions of the 11 ST clusters.

(C and D) Normalized enrichment scores (NESs) of proliferative (C) and immune (D) MSigDB cancer hallmark pathways across ST clusters.

(E) Cox regression for each ST cluster identifies survival-associated clusters (hazard ratios on a log scale; error bars indicate 95% confidence intervals; $p < 0.05$).

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to group ~6.7 million pseudo-spots into 7,295 spatially coherent domains (average 6.6 per slide), representing spatially contiguous regions with similar transcriptomic profiles. Gene expression was averaged within each domain, and domain profiles were jointly clustered to identify 11 spatial transcriptomic clusters (ST clusters) shared across patients, providing a consistent cross-tumor reference (Figures 5A and 5B). We validated this approach using the Bassiouni et al. ST dataset, achieving Rand and adjusted Rand indices of 0.75 and 0.49 (Figure S6A). The ST clusters exhibited subtype-specific patterns ($p = 0.003$; Figure S6B). Further analysis revealed distinct patterns in hallmark pathway activity, cell-type composition, oncogene and tumor suppressor expression, and cell-cell interactions (Figures 5C and 5D; Figures S6C–S6E). Notably, a higher abundance of cluster 4 was associated with worse disease-free survival, while cluster 5 was associated with better survival ($p = 0.02$ and 0.03 ; Figure 5E).

To stratify TCGA breast tumors based on spatial architecture, we performed hierarchical clustering on tumor profiles defined by the 11 ST clusters. Multiple metrics (elbow, Davies-Bouldin, and PAC) identified five patient-level meta-clusters as optimal (Figure S6F). Three meta-clusters comprising 98% of patients (all but 12) showed distinct spatial TME signatures, and we refer to these as “SpatioTypes” (Figure 5F).

Each SpatioType exhibits a distinct TME composition. The proliferation-enriched SpatioType is composed of ST clusters 9 and 11 (Figures 5C and 5F). The immune-modulated SpatioType features elevated immune activity, with cluster 5—one of two clusters nominally associated with disease-free survival—linked to favorable outcomes (Figure 5E). By contrast, the immune-inactive SpatioType is dominated by clusters 2 and 4, marked by low immune activation (Figures 5D and 5F). Cluster 2 is characterized by high cancer cell density, oncogene upregulation, and tumor suppressor downregulation, while cluster 4 is the other cluster associated with poor disease-free survival ($p = 0.02$, FDR = 0.14; Figure 5E).

Consistent with these spatial patterns, TCGA bulk RNA sequencing (RNA-seq) data reflected the molecular profiles of the SpatioTypes: proliferation-enriched patients showed strong activation of proliferative hallmark pathways, while immune-modulated and immune-inactive patients showed high and low immune pathway activation, respectively (Figure S6G). TCGA tumor-infiltrating lymphocyte (TIL) maps³⁷ showed TIL proportions were 24% higher in immune-modulated versus immune-inactive tumors ($p = 0.04$; Figure S6H), while the proliferation-enriched SpatioType was associated with higher pathological stage (>1) ($p = 0.006$; Figure S6I), consistent with their immune and proliferative phenotypes, respectively.

Survival analysis of the 853 patients with both survival and slide data revealed that these three spatial SpatioTypes

markedly stratify disease-free survival ($p = 0.008$, two-sided log-rank test). Significant differences in survival were observed between immune-modulated and immune-inactive patients ($p = 0.001$, two-sided log-rank test; Figure 5G). Notably, the immune-inactive SpatioType ($n = 218$) was associated with poorer prognosis, even after adjusting for age and tumor stage in the multivariate Cox proportional hazards model (HR = 2.04, $p = 0.002$, two-sided Wald test). This stratification extended across clinical subtypes, including HR+, HER2+, and TNBC (Figure 5H).

To validate these findings, we applied Path2Space to METABRIC³⁸ patients ($n = 141$) and assigned tumors to TCGA-derived SpatioTypes (STAR Methods; Figure S7A). The resulting SpatioType assignments recapitulated the survival stratification observed in TCGA ($p = 0.01$, log-rank test; Figure S7B), with the immune-inactive SpatioType remaining significantly associated with poor prognosis after adjusting for tumor stage and age (HR = 5.45, $p < 0.001$). These trends were consistent across clinical subtypes (Figure S7C). SpatioTypes consistently stratified survival within the poor Nottingham Prognostic Index (NPI) subgroup ($p = 0.005$; Figure S7D) and provided independent prognostic value beyond NPI (likelihood ratio test $p = 0.02$).

Inferred spatial heterogeneity of HER2 predicts trastuzumab response

To quantify spatial heterogeneity in Path2Space-inferred gene expression, we defined spatial neighborhood diverseness (SPAND), which measures local expression variability within $100 \times 100 \mu\text{m}$ neighborhoods across a slide, with higher values indicating greater heterogeneity (STAR Methods; Figure 6A).

We next evaluated whether HER2 SPAND predicts trastuzumab response. Given that HER2-high cells are more sensitive to trastuzumab-mediated immune response,^{39–44} we hypothesized that tumors with high SPAND, where HER2-high cells intermix with HER2-low cells, would show a better response, as an immune response elicited by targeted HER2-high cells could influence nearby HER2-low cells. We applied Path2Space to four trastuzumab-treated breast cancer cohorts (TransNEO, PBCP, IMPRESS, and Cedars-Sinai) and computed HER2 pathway activity SPAND scores normalized by cancer cell fraction (HER2 per cancer).

HER2 SPAND predicted pathological complete response (pCR) across cohorts: TransNEO (AUC = 0.80), PBCP (AUC = 0.69), IMPRESS (AUC = 0.72), and Cedars-Sinai (AUC = 0.83) (Figures 6B–6E), outperforming mean HER2 per cancer (Figure S7E) and top ST prediction models (Figure S7F). Predictive performance of HER2 SPAND was specific to trastuzumab, with a weaker association with chemotherapy response (AUC = 0.61 and 0.57 in TransNEO and IMPRESS). HER2 SPAND also outperformed bulk RNA-seq HER2 expression in TransNEO and matched it in PBCP (Figures 6B and 6C). Performance

(F) Hierarchical clustering of ST cluster abundances across TCGA samples defines three SpatioTypes. Heatmap shows patients (rows) and cluster proportions (columns), defining proliferation-enriched, immune-modulated, and immune-inactive SpatioTypes.

(G) Kaplan-Meier disease-free survival across SpatioTypes ($p = 0.008$, two-sided log-rank test), with poorer outcomes for the immune-inactive group, particularly versus the immune-modulated group ($p = 0.001$, two-sided log-rank test).

(H) Kaplan-Meier disease-free survival stratified by molecular subtype (HR+, HER2+, and TNBC). SpatioType colors correspond to (F) and (G). Statistical significance assessed using one-sided log-rank tests.

See also Figures S6 and S7

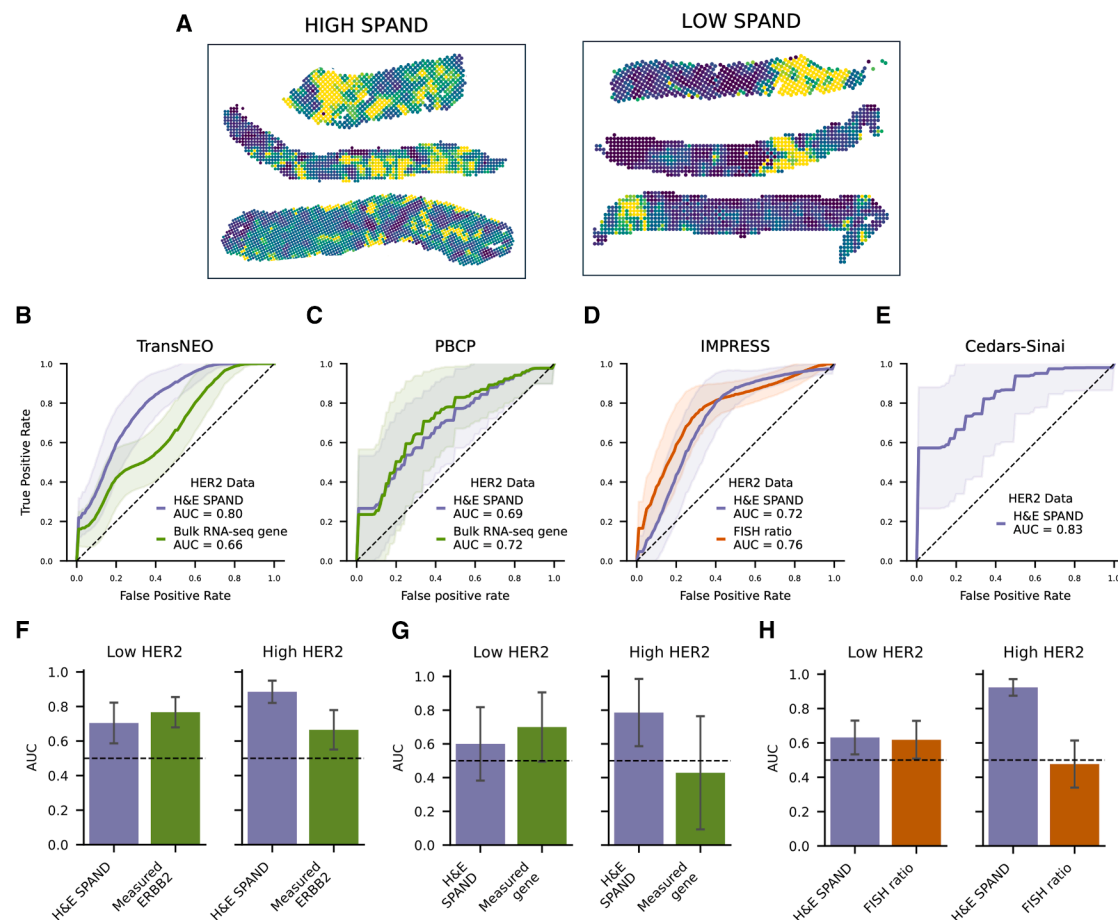


Figure 6. Spatial expression heterogeneity predicts response to trastuzumab

(A) Representative TransNEO slides with high (left) versus low (right) HER2 SPAND scores.

(B–E) ROC curves evaluating HER2 SPAND for predicting pCR to trastuzumab in TransNEO (B), PBCP (C), IMPRESS (D), and Cedars-Sinai (E). Where available, HER2 SPAND is compared with molecular HER2 measurements: bulk RNA-seq *ERBB2* expression (B, C) or FISH HER2/CEP17 ratios (D). Shaded regions indicate bootstrap standard errors ($n = 1,000$).

(F–H) Trastuzumab response prediction (AUC) of HER2 SPAND compared with molecular HER2 assays, stratified by HER2 level. In TransNEO (F) and PBCP (G), HER2 SPAND is compared with bulk RNA-seq *ERBB2* expression, with low and high groups defined by cohort-specific medians. In IMPRESS (H), HER2 SPAND is compared with FISH HER2/CEP17 ratios, stratified by median amplification. Bars show AUC values, and error bars indicate standard errors (1,000 bootstrap iterations).

See also Figure S7

was strongest in HER2-high tumors, where SPAND substantially outperformed bulk RNA-seq (TransNEO: 0.89 vs. 0.65; PBCP: 0.78 vs. 0.40; Figures 6F and 6G). In IMPRESS, HER2 SPAND was comparable to HER2/CEP17 FISH (Figure 6D) but more effective in high-amplification tumors (0.92 vs. 0.55; Figure 6H). In contrast to prior non-spatial measures linking HER2 heterogeneity to poor trastuzumab response,^{45,46} SPAND captures spatially resolved heterogeneity at micro-meter-scale neighborhoods.

Spatial clustering predicts response to trastuzumab and chemotherapy

To identify spatial biomarkers of treatment response, we used the 11 ST clusters derived from TCGA to represent tumor composition across cohorts. Using Path2Space-inferred

spatial domains, we quantified cluster proportions in TransNEO, PBCP, and IMPRESS tumors. We trained a logistic regression model on 11-cluster composition in chemotherapy-treated TransNEO patients. In 5-fold cross-validation, Path2Space achieved an AUC of 0.75, outperforming a direct image-based model and matching an H&E-imputed bulk expression approach (Figure 7A; Table S4; STAR Methods). In independent cohorts, Path2Space generalized well, achieving AUC = 0.89 in PBCP and 0.74 in IMPRESS chemotherapy-treated TNBC patients, where alternative models failed to generalize (Figure 7A; Table S4). Path2Space also outperformed five published H&E-based predictors evaluated on IMPRESS, including Huang et al.²⁹ (Figure 7A).

We next applied the same representation to predict trastuzumab response. We trained a logistic regression model in

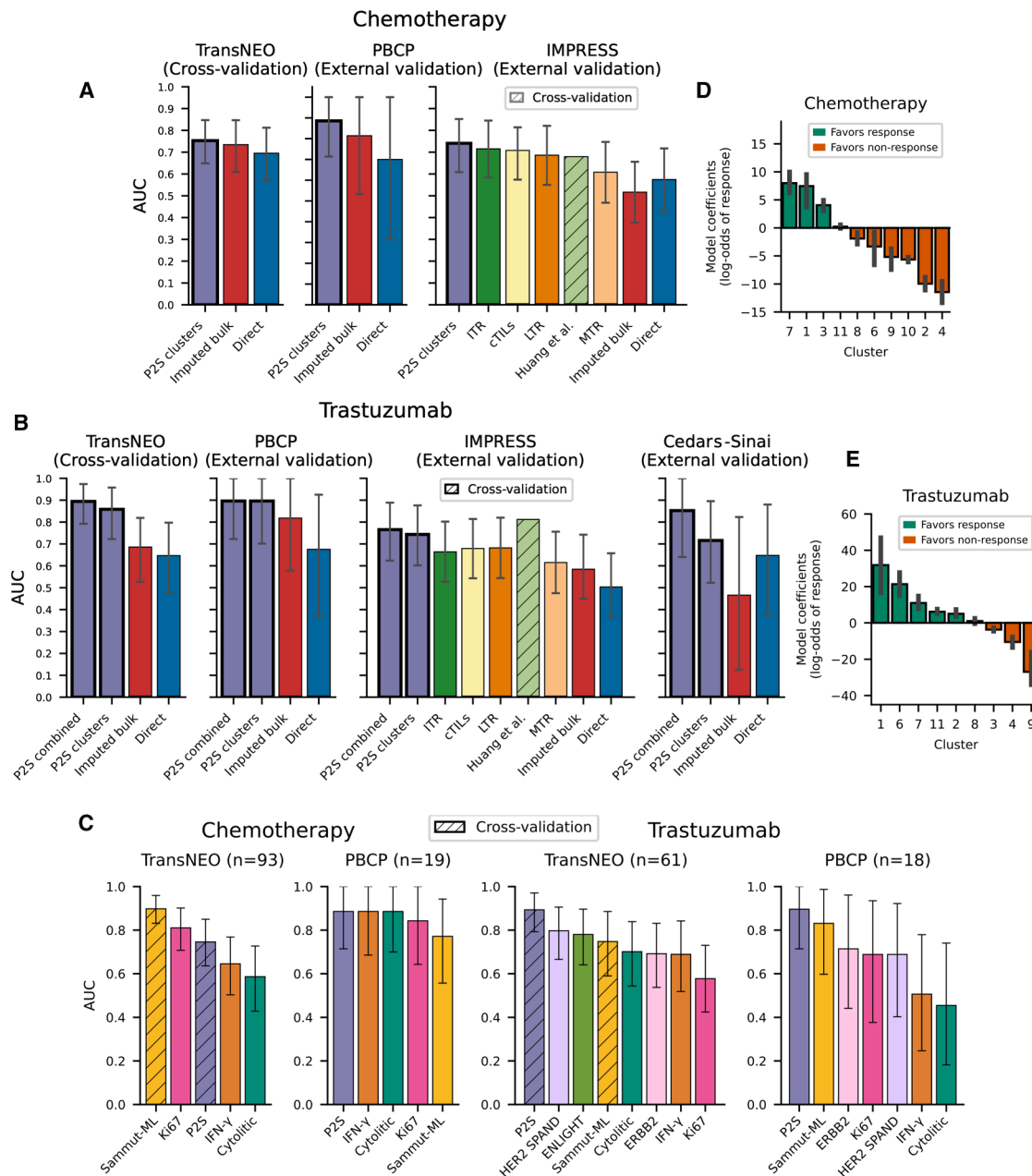


Figure 7. Path2Space predicts chemotherapy and trastuzumab response from TME composition

(A and B) Response prediction (AUC) for chemotherapy (A) and trastuzumab (B) in TransNEO (cross-validation), PBCP and IMPRESS (external validation), and Cedars-Sinai (external validation; trastuzumab only). Bar plots compare Path2Space with H&E-based models (imputed bulk and direct). For chemotherapy, the Path2Space spatial cluster-based model (P2S clusters) is shown. For trastuzumab, both the cluster-only model and a combined model integrating spatial clusters with per-cancer HER2 SPAND (P2S combined) are shown. In IMPRESS, we additionally report Huang et al. cross-validation performance and H&E-based predictors from Aswolinskiy et al. lymphocyte-tumor ratio (LTR), computational TIL score (cTILs), inflamed tumor ratio (ITR), and mitoses-tumor ratio (MTR). Error bars denote 95% confidence intervals (1,000 bootstrap iterations), and bar height indicates AUC.

(C) Comparison of Path2Space (P2S clusters for chemotherapy; P2S combined and HER2 SPAND for trastuzumab) with bulk transcriptomic and multimodal predictors (Sammur-ML, ENLIGHT, *ERBB2* expression, Ki-67, cytolytic activity score, IFN- γ signature) for chemotherapy (left) and trastuzumab (right) in the TransNEO and PBCP cohorts. Error bars show 95% confidence intervals (1,000 bootstrap iterations).

(D and E) Spatial cluster contributions in logistic regression models for chemotherapy (D) and trastuzumab (E) response. Bars show mean coefficients across 5-fold cross-validation with standard errors, and positive (green) and negative (orange) coefficients indicate association with response and non-response.

See also Figure S7

trastuzumab-treated TransNEO patients, achieving an AUC of 0.86 in cross-validation. In external validation, the model achieved AUCs of 0.90 in PBCP, 0.74 in IMPRESS, and 0.72 in Cedars-Sinai, outperforming models based on direct image features or H&E-imputed bulk expression (Figure 7B). Integration with HER2 SPAND further improved performance, yielding AUCs of 0.89, 0.90, and 0.77 in TransNEO, PBCP, and IMPRESS, respectively (Figure 7B). In the validation cohorts, false discovery among predicted responders decreased with increasing prediction score, with lower scores indicating less reliable predictions (Figure S7G). Applying a predefined threshold to these cohorts, patients above the threshold showed higher odds of response (chemotherapy odds ratio [OR] = 3.38, $p = 0.008$; trastuzumab OR = 2.57, $p = 0.03$; Figure S7H).

We compared our models to published approaches using non-spatial molecular data, all trained and evaluated on the same cohorts. For chemotherapy, Sammut-ML²⁸ outperformed Path2Space in cross-validation, but Path2Space generalized better to the external validation cohort PBCP (Figure 7C). For trastuzumab, Path2Space outperformed Sammut-ML in cross-validation on TransNEO (AUC = 0.75) and in external validation on PBCP (AUC = 0.83) and outperformed ENLIGHT⁴⁷ (AUC = 0.78) (Figure 7C). Path2Space also matched or outperformed bulk RNA-seq-based biomarkers across cohorts (Figure 7C).

To interpret model features, we analyzed logistic regression coefficients. Cluster 4 showed the strongest association with poor chemotherapy response, followed by cluster 2 (Figure 7D), consistent with TCGA, where both define the poor-prognosis immune-inactive SpatioType, and cluster 4 shows the strongest association with poor survival (Figures 5E and 5F). In the trastuzumab model, cluster 4 remained associated with poor response (Figure 7E), albeit less strongly. Clusters 1 and 6, associated with better response, showed elevated HER2 SPAND levels, with cluster 6 among the highest (Figure S7I).

DISCUSSION

Path2Space provides a practical and scalable implementation that enables large-scale spatial analyses. Such analyses could in principle be conducted using other prediction frameworks, and we expect that this study will motivate others to develop improved approaches in this line of biomarker research.

We investigated trastuzumab response using two complementary approaches. First, using HER2 SPAND, we found that tumors with high SPAND—indicating intermixed HER2-high and HER2-low regions—showed better response than low SPAND tumors. This is consistent with trastuzumab's Fc receptor-mediated immune cytotoxicity³⁹ and complements prior studies of HER2 heterogeneity.^{45,46} Second, we represented tumors by their spatial transcriptomic cluster composition using 11 TCGA-derived clusters. This compact representation enables simple models to predict treatment response directly from H&E slides, highlighting spatial environments associated with clinical outcomes.

Path2Space supports two complementary clinical applications. First, H&E-based analysis enables identification of spatial features associated with clinical outcomes in large retrospective

cohorts lacking molecular profiling. These biomarkers can then be validated using targeted assays in prospective settings. Second, Path2Space enables direct prediction from routine H&E slides, reducing reliance on costly molecular testing and yielding image-derived biomarkers from whole-slide histology.⁴⁸ Stratifying patients using Path2Space-based biomarkers requires only deployable computational infrastructure, positioning these biomarkers as a scalable, cost-efficient alternative to molecular profiling.

Finally, the framework is adaptable to other cancer types and spatial omics modalities. Spatial proteomics may be particularly valuable where post-translational regulation decouples protein from mRNA levels, as demonstrated for estrogen and progesterone receptors in breast cancer.⁴⁹ Spatial methylation profiling could enable the discovery of epigenetic biomarkers. For example, *MGMT* promoter methylation predicts temozolomide response in glioblastoma,⁵⁰ yet its spatial distribution remains unexplored and may reveal heterogeneity relevant to treatment resistance.

Limitations of the study

We focus on H&E because such cohorts are larger, more accessible, and less costly than emerging spatial omics datasets. We leverage Visium data, where H&E staining is integral to the workflow, ensuring matched image-expression pairs. By contrast, fluorescence-based platforms (e.g., CosMx, Xenium, and spatial proteomics) do not require H&E imaging, and paired H&E data are not routinely generated, limiting matched training data.^{51,52} As ST technologies improve in resolution and depth, we anticipate corresponding gains in prediction accuracy and cell-type inference. However, the current architecture is optimized for Visium-resolution data (~55 μm spots). Extending Path2Space to single-cell or subcellular platforms will require architectural modifications as higher-resolution training data become available. A related limitation concerns cell-type inference granularity. While Path2Space enables estimation of major cell-type compositions validated against pathologist annotations (Figures 4C and 4D), finer cell-type resolution is an important consideration for future development. When we evaluated fine-grained deconvolution using Xenium-derived ground-truth labels from Janesick et al.,⁵³ Path2Space achieved high accuracy for abundant cell types (tumor AUC = 0.86, CD4 T cells AUC = 0.82, and stromal AUC = 0.79), but performance declined for less abundant populations (Figure S5E). Importantly, deconvolution based on measured ST data exhibited the same abundance-dependent decline (Figure S5F), underscoring that rare cell types are inherently challenging to infer at spot-level resolution, even from measured expression data. Both approaches showed strongly correlated performance across cell types ($r = 0.83$; Figures S5E and S5F), indicating that Path2Space's reduced accuracy for low-abundance cell types reflects intrinsic constraints of Visium-resolution data (~1–10 cells per spot) rather than model-specific shortcomings. Nevertheless, the broad cell-type estimates achieved here were sufficient to characterize the ST clusters and SpatioTypes underlying our biomarker analyses. Capturing rare and specialized populations would benefit from higher-resolution training data, a

natural extension as training data from platforms such as Visium HD become more widely available.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Eytan Ruppin (eytan.ruppin@cshs.org).

Materials availability

This study did not generate any new, unique reagents. This is a computational study using publicly available datasets.

Data and code availability

The source code for Path2Space is freely available for non-commercial academic research through Zenodo (DOI: [10.5281/zenodo.14729336](https://doi.org/10.5281/zenodo.14729336)). Comprehensive tutorials and example input and output files are provided to facilitate implementation and usage of the software.

This study analyzed existing, publicly available datasets. Accession numbers and links are listed in the [key resources table](#).

- Bassiouni et al. Visium cohort: GEO: GSE210616.
- HER2+ dataset: Zenodo: <https://doi.org/10.5281/zenodo.4751624>.
- HEST: <https://huggingface.co/datasets/MahmoodLab/hest>.
- HTAN: <https://humantumoratlas.org/>.
- Martinez et al.: GEO: GSE213688.
- PanoptiLS: <https://sites.google.com/view/panoptils/>.
- TCGA-BRCA: <https://portal.gdc.cancer.gov/> (TCGA Research Network).
- METABRIC: EGA: EGAS00000000083.
- TransNEO and PBCP: H&E images available at Zenodo (<https://zenodo.org/record/6337925>) and patient response data at EGA: EGAS00001004582.
- IMPRESS: https://drive.google.com/drive/folders/1fNf-F_aplm6ACJTWO1vGqbb-DdaP4K_r.
- Cedars-Sinai: available from the [lead contact](#) upon reasonable request.

Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, E.D.S., E.M.C., J.S.L., J.Y., and E.R.; methodology, E.D.S., E.M.C., P.J., and J.S.L.; software, E.D.S., E.M.C., Y.C., C.W., and J.S.L.; formal analysis, E.D.S., E.M.C., R.L., Y.C., A.S., T.C., B.R., T.-G.C., S.B., S.R.D., S. Patiyal, S. Patkar, A.W., R.K.B., C.W., R.P., S.C.K., E.P., J.Y., and J.S.L.; data curation, E.D.S., E.M.C., S.-J.S., and C.C.; investigation, S.-J.S., C.C., S.L., P.S.R., S.R.V.K., D.-T.H., A.P., Y.C., and C.W.; validation, E.D.S., E.M.C., E.P., T.H., M.P.N., K.A., Y.C., and C.W.; visualization, E.D.S. and E.M.C.; resources, Y.Y. and K.S.; writing – original draft, E.D.S. and E.M.C.; writing – review & editing, all authors; supervision, E.D.S., E.R., E.P., J.Y., and J.S.L.; funding acquisition, E.R.

DECLARATION OF INTERESTS

E.D.S., E.M.C., and E.R. are listed as inventors on a provisional patent (application no. 63/703,060, United States, 2024) filed based on the methodology outlined in this study. E.R. is a (non-paid) member of the scientific advisory boards of Pangea Biomed (divested), GSK Oncology, and the ProCan project. E.R. is a founder of MedAware Ltd.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT to improve the readability and language of the work. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Path2Space source code	This paper; https://zenodo.org/records/14729336	10.5281/zenodo.14729336
Bassiouni et al. Visium cohort	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE210616	GEO: GSE210616
HER2+ dataset	https://zenodo.org/records/4751624	10.5281/zenodo.4751624
HEST	https://huggingface.co/datasets/MahmoodLab/hest	N/A
HTAN	https://humantumoratlas.org/	N/A
Martinez et al.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE213688	GEO: GSE213688
TCGA-BRCA	https://portal.gdc.cancer.gov/	TCGA Research Network
METABRIC	https://ega-archive.org/studies/EGAS00000000083	EGAS00000000083
PanopTILs dataset	https://sites.google.com/view/panoptils/	N/A
TransNEO and PBCP H&E images	https://zenodo.org/record/6337925	10.5281/zenodo.6337925
TransNEO and PBCP clinical data	https://ega-archive.org/studies/EGAS00001004582	EGAS00001004582
IMPRESS	https://drive.google.com/drive/folders/1fNf-F_aplm6ACJTW01vGqbb-DdaP4K_r	N/A
Cedars-Sinai trastuzumab cohort	Available from lead contact upon request	N/A
Software and algorithms		
Path2Space	This paper; https://zenodo.org/records/14729336	10.5281/zenodo.14729336
Python (3.10.8)	https://python.org	RRID: SCR_008394
R (4.3.2)	https://www.r-project.org/	RRID: SCR_001905
NumPy (1.24.4)	https://numpy.org	RRID: SCR_008633
Pandas (2.2.2)	https://pandas.pydata.org/	RRID: SCR_018214
SciPy	https://scipy.org/	RRID: SCR_008058
scikit-learn (1.5.1)	https://scikit-learn.org/stable/	RRID: SCR_002577
Scanpy (1.10.4)	https://scanpy.readthedocs.io/	RRID: SCR_018139
lifelines (0.28.0)	https://lifelines.readthedocs.io/en/latest/	RRID: SCR_024899
PyTorch (2.4.1)	https://pytorch.org/	RRID: SCR_018536
torchvision	https://pytorch.org/vision/stable	N/A
timmm (1.0.15)	https://huggingface.co/docs/timmm/en/index	N/A
OpenSlide (1.3.1)	https://openslide.org/	RRID: SCR_016672
OpenCV (4.6.0)	https://opencv.org	RRID: SCR_015526
Pillow (9.2.0)	https://pillow.readthedocs.io/en/stable/	RRID: SCR_023337
Matplotlib (3.7.2)	https://matplotlib.org/	RRID: SCR_008624
Seaborn (0.13.2)	https://seaborn.pydata.org/	RRID: SCR_018132
ggplot2 (3.5.1)	https://ggplot2.tidyverse.org/	RRID: SCR_014601
StainTools (2.1.2)	https://github.com/Peter554/StainTools	N/A
CellPose	https://github.com/MouseLand/cellpose	N/A
Squidpy	https://squidpy.readthedocs.io/	N/A
GSEAPY	https://gseapy.readthedocs.io/	N/A
PySAL (24.01)	https://pysal.org/	N/A
SpaGCN	https://github.com/jianhuupenn/SpaGCN	N/A
SpaCET	https://github.com/data2intelligence/SpaCET	N/A
Seurat	https://satijalab.org/seurat/	RRID: SCR_016341

(Continued on next page)

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
survival R package (3.5.7)	https://cran.r-project.org/package=survival	RRID: SCR_021137
ConsensusClusterPlus (1.62.0)	https://bioconductor.org/packages/ConsensusClusterPlus	RRID: SCR_016954
factoextra (1.0.7)	https://cran.r-project.org/package=factoextra	N/A
cluster R package (2.1.4)	https://cran.r-project.org/package=cluster	N/A
compositions R package (2.0–6)	https://cran.r-project.org/package=compositions	N/A
SpatialDM	https://github.com/StatBiomed/SpatialDM	N/A
Trident	https://github.com/mahmoodlab/TRIDENT	N/A
CTransPath	https://github.com/Xiyue-Wang/TransPath	N/A
GigaPath	https://github.com/prov-gigapath/prov-gigapath	N/A
H-Optimus-1	https://huggingface.co/bioptimus/H-optimus-1	N/A
Virchow2	https://huggingface.co/paige-ai/Virchow2	N/A
Phikon-v2	https://huggingface.co/owkin/phikon-v2	N/A
UNI2-h	https://huggingface.co/MahmoodLab/UNI2-h	N/A
HoVer-Net	https://github.com/vqdang/hover_net	N/A
ST-Net	https://github.com/bryanhe/ST-Net	N/A
BLEEP	https://github.com/bowang-lab/BLEEP	N/A
STco	https://github.com/shizhiceng/STco	N/A
mclSTExp	https://github.com/shizhiceng/mclSTExp	N/A
EGNv2	https://github.com/Yan98/EGN/tree/main/v2	N/A
EGGN	https://github.com/Yan98/EGN	N/A
DeepPT	https://doi.org/10.5281/zenodo.11125591	N/A
HisToGene	https://github.com/maxpmx/HisToGene	N/A
HistoSPACE	https://github.com/samrat-lab/HistoSPACE	N/A
TRIPLEX	https://github.com/NEXGEM/TRIPLEX	N/A
STFlow	https://github.com/Graph-and-Geometric-Learning/STFlow	N/A
DeepSpaCE	https://github.com/tmonjo/DeepSpaCE	N/A
SEPAL	https://github.com/BCV-Uniandes/SEPAL	N/A
STImage	https://github.com/BiomedicalMachineLearning/STImage	N/A
Hist2ST	https://github.com/biomed-AI/Hist2ST	N/A
LocalNet (ViT-Base-Patch16-224)	https://huggingface.co/timm/vit_base_patch16_224.augreg_in21k	N/A
Other		
MSigDB Cancer Hallmark gene sets	https://www.gsea-msigdb.org/	N/A
WikiPathways ErbB Signaling (WP673)	https://www.wikipathways.org/	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human cohort descriptions

This is a computational study; no experimental models were directly used. All analyses were performed on publicly available, de-identified human tissue data from prior studies with existing ethics approvals and informed consent. The datasets used are described below.

Spatial transcriptomics cohorts

External validation was performed on three independent spatial transcriptomics cohorts encompassing both fresh frozen and FFPE tissue preservation methods. For HEST, all 5 samples from 4 patients passed QC. For Martinez *et al.*, analysis was limited to the sub-set of cancer specimens with full-resolution H&E images available (n = 5); four samples from four patients passed QC. For HTAN, 9 of 44 samples (from 7 patients) passed QC thresholds, reflecting comparatively lower data quality in this cohort. The Bassiouni *et al.* training cohort retained 22 of 43 tissue sections from 14 patients after QC filtering. QC criteria are detailed in the Quality control criteria section.

Cell type annotation cohort

The PanopTILs dataset comprises H&E images and cell type annotations of 1,709 regions of interest (ROIs) from 151 TCGA breast cancer patients. This dataset was used to train the cell type fraction estimation module. External validation was performed using pathologist-annotated HEST samples.

Application cohorts

TCGA-BRCA: The Cancer Genome Atlas breast cancer cohort includes 1,096 samples from 976 patients, with matched diagnostic H&E whole-slide images, bulk RNA-seq, and clinical outcomes including disease-free survival. Data were obtained through the Genomic Data Commons (GDC) portal.

METABRIC: The Molecular Taxonomy of Breast Cancer International Consortium cohort includes 158 breast cancer patients with H&E images and clinical outcome data (141 retained after filtering), used for external validation of SpatioType survival associations.

TransNEO: A prospective clinical cohort of HER2+ (n = 61) and HER2– (n = 93) breast cancer patients treated with neoadjuvant chemotherapy (HER2– patients) or chemotherapy plus trastuzumab (HER2+ patients), with pCR outcomes and pre-treatment H&E slides.

PBCP: The Personalised Breast Cancer Programme cohort includes HER2+ (n = 18) and HER2– (n = 19) breast cancer patients with treatment response data and pre-treatment H&E slides, used for external validation of treatment response prediction.

IMPRESS: A clinical cohort of HER2+ (n = 62) and HER2– (n = 64) breast cancer patients with treatment response data, HER2/CEP17 FISH measurements, and H&E slide images.

Cedars-Sinai: A clinical cohort of HER2+ breast cancer patients (n = 26) treated with trastuzumab, with treatment response data and H&E slides, used for external validation of trastuzumab response prediction.

Ethics statement

All human data were obtained from publicly available repositories with existing institutional review board approvals and informed consent from original studies. TCGA data use is governed by NIH Genomic Data Sharing Policy. For METABRIC, TransNEO, and IMPRESS cohorts, data access followed the data use agreements established by the respective consortia. No additional ethics approval was required for this computational analysis of existing de-identified data.

METHOD DETAILS

Gene expression data preprocessing

Gene filtering

Due to the high proportion of missing values typical in spatial transcriptomics data, we focused on 14,068 genes that were detected in at least 5% of spots on each slide in the Bassiouni *et al.* dataset, ensuring sufficient data for robust training across genes.

Gene expression normalization

To predict spatial gene expression from histopathology, we used $\log_{10}$ -transformed raw counts ($\log_{10}(\text{count} + 1)$) as target labels for model training and evaluation, rather than library size-normalized values. This choice was guided by a direct comparison with a Path2Space model trained to predict $\log_{10}(\text{CPM} + 1)$ values using the same Bassiouni *et al.* dataset, model architecture, and training pipeline. The model trained on raw counts outperformed the CPM-based model, predicting a substantially greater number of genes with $\text{PCC} > 0.4$ between predicted and observed values. Specifically, while the CPM model yielded only 1,540 such genes in cross-validation, the raw count-based model identified 6,629 genes exceeding this threshold. The count model captured 1,536 of the 1,540 (>99.7%) CPM-predicted genes.

To further assess the biological relevance of the ST predictions generated by the two models, we evaluated their ability to infer cellular composition in annotated tumor regions from TCGA breast cancer slides using the PanopTILs dataset (as in Figure S5B). We applied SpaCET deconvolution with scRNA-seq-derived reference signatures to the gene expression values predicted by each model. Predictions based on log-transformed raw counts showed markedly higher concordance with PanopTILs-derived annotations, achieving mean correlations of 0.69 (cancer), 0.57 (lymphocytes), and 0.40 (stromal cells). In contrast, predictions based on CPM values yielded significantly lower correlations of 0.40, 0.24, and 0.16, respectively (two-sided Mann–Whitney U-test, $p < 0.01$ for all comparisons; Figure S1B).

Repeating this analysis on tissue sections with cell type annotations from the HEST cohort (as in Figure 4D) confirmed that the counts model significantly outperformed the CPM model for lymphocyte classification (AUC = 0.92 vs. 0.59; two-sided DeLong test, $p < 0.01$), and maintained higher performance for cancer and stromal cells ($p = 0.34$ and $p = 0.26$, respectively; Figure S1C). These results highlight the advantage of using log-transformed raw counts as predictive targets for capturing biologically meaningful spatial gene expression patterns from histopathology.

Path2Space architecture

The architecture consists of four main components (Figure 1A):

Image preprocessing

For image preprocessing, we used the highest available resolution for each ST dataset—typically 40× magnification, corresponding to approximately 0.35 microns per pixel. For each spot, we extracted a square tile centered on the spot with a side length of 80 microns.

To identify tissue-containing regions within each slide, we applied Sobel edge detection to compute the weighted gradient magnitude. Tiles were excluded if more than 50% of their pixels had a gradient magnitude below 0.15, indicating insufficient tissue content. To reduce staining variability and batch effects, we applied Macenko's color normalization⁵⁴ method to the retained tiles.

For prediction on TCGA and clinical cohorts, we similarly used the highest available magnification (40× for most TCGA slides and 20× for clinical cohorts), applied the same tissue filtering and color normalization steps, and extracted 80-micron square tiles to serve as pseudo-spots. The number of pixels per tile was adjusted based on image resolution to ensure consistent physical size across datasets.

Feature extraction

Tile images corresponding to spots were processed using CTransPath,¹⁹ a foundational digital pathology model trained through self-supervision on histopathology images. Each tile was represented by a 768-dimensional CTransPath feature vector.

To evaluate feature representation impact, we trained Path2Space using features from five recent pathology foundation models: GigaPath, Virchow2, Phikon-v2, UNI2-h, and H-Optimus-1, with identical architectures and only input dimensions adjusted. In cross-validation, GigaPath, H-Optimus-1, and CTransPath showed similar performance with no significant differences (median PCC ~0.5). On external validation, performance varied by cohort, with CTransPath achieving highest correlation on HEST while other models performed comparably or slightly better on Martinez and HTAN. Averaged across external cohorts, H-Optimus-1 achieved median PCC of 0.37 versus 0.36 for CTransPath, demonstrating that CTransPath remains competitive with state-of-the-art encoders developed after Path2Space (Figure S1A).

MLP regression

This component built a predictive model linking CTransPath features to whole-genome spot-level gene expression. The model architecture includes three layers: (1) an input layer with 768 nodes, corresponding to the size of the CTransPath feature vector; (2) a hidden layer with 768 nodes; and (3) an output layer with 14,068 nodes, each representing a gene. ReLU activation is applied to each layer to introduce non-linearity and enhance model performance.

Spatial smoothing

Gene expression values were smoothed by averaging each spot's expression with that of all its neighboring spots within a 200-micrometer radius, the minimum distance between spots in the dataset. On average, this resulted in 8 neighbors per spot being used for smoothing. This procedure was applied to both model predictions and original expression measurements (Figure S1F) and was used for model evaluation and downstream analyses.

To quantify the contribution of spatial smoothing, we compared model performance with and without smoothing applied to predictions, using unsmoothed measured expression as ground truth in both cases. Spatial smoothing substantially improved prediction accuracy: in cross-validation, 1,376 genes achieved PCC > 0.4 with smoothing compared to 798 without; in external validation, 498 genes exceeded this threshold with smoothing versus only 51 without (Figure S1G; note that Figure 2 reports higher numbers because both predictions and measurements are smoothed there). Importantly, this improvement extends to downstream biological applications. Applying SpaCET deconvolution to Path2Space predictions on two HEST samples with spot-level pathologist annotations, spatial smoothing significantly improved cell type classification for all three major cell types: TILs, stromal cells, and cancer cells ($p < 0.001$ for all comparisons, two-sided DeLong test; Figure S1H). These results demonstrate that spatial smoothing enhances both molecular prediction accuracy and biological interpretability.

Model training

Model optimization

Models were trained using stochastic gradient descent with mini-batch processing. Optimization was performed using the Adam optimizer with an initial learning rate of 1×10^{-4} . Mean squared error (MSE) between predicted and measured values was used as the loss function.

Training of the MLP regressors and classifiers was conducted for up to 500 epochs. To reduce overfitting, a dropout rate of 20% was applied to the first hidden layer. Early stopping was implemented to improve computational efficiency and further mitigate overfitting: training was terminated if no improvement was observed in the mean correlation between predicted and measured gene expression values on the validation set for 50 consecutive epochs.

Cross-validation procedure

Our training and evaluation strategy followed the protocol established in our previous work^{8,9} and was based on a leave-one-patient-out cross-validation scheme with ensemble prediction. Specifically, we performed 22 iterations, each corresponding to holding out all slides from one patient as an independent test set.

For each iteration, models were trained using data from the remaining patients. Within this training set, we created seven independent training-validation splits by holding out all slides from a single, distinct patient for validation, while using the remaining patients for training. Validation performance was used for early stopping.

At all stages, slides from a given patient were assigned exclusively to a single role (training, validation, or testing), ensuring the absence of patient-level data leakage.

Ensemble prediction strategy

As a result, for each of the 22 cross-validation test patients, seven models were trained, each corresponding to a different held-out validation patient. Final predictions for the held-out test patient were obtained by averaging the predictions across these seven models. This cross-validation training procedure resulted in 154 trained models (7 models for each of the 22 test patients). For external validation datasets and clinical H&E cohorts, predictions from all 154 models were averaged to produce final predictions.

In a post-hoc analysis comparing this ensemble strategy to alternative approaches, we found that ensemble averaging significantly enhanced predictive performance compared to using the single best-performing model from the validation set. In cross-validation, ensemble averaging predicted 464 more genes with $PCC > 0.4$ than the best single model (6,629 vs. 6,165 genes; median PCC : 0.39 vs. 0.36, $p < 0.0001$, two-sided Mann–Whitney U-test). In external validation, the ensemble approach achieved similar improvements, predicting 1,124 more genes with $PCC > 0.4$ (5,453 vs. 4,329 genes; median PCC : 0.36 vs. 0.34, $p < 0.0001$) (Figure S1D). Notably, the ensemble approach identified 25% more robustly predicted genes that achieved $PCC > 0.4$ in both cross-validation and external validation cohorts (4,807 vs. 3,835 genes), demonstrating improved generalizability (Figure S1E).

Training data requirements

To evaluate how Path2Space performance scales with training data size, we trained models on progressively larger subsets of the Bassiouni *et al.*²⁰ cohort (3–11 patients; one sample per patient; each restricted to 1,000 spots to control for sample-size variation) to predict the 988 HVGs (same ones used in the benchmarking analysis) and evaluate prediction generalization on three independent validation cohorts—HEST,²⁵ HTAN,²⁷ and Martinez *et al.*²⁶ Performance saturated rapidly as training size was increased in all validation cohorts. In HEST, accuracy showed minimal improvement beyond training on 5 patients / 5,000 spots from Bassiouni *et al.* (median PCC = 0.43 vs. 0.44 for the 11-patient model). HTAN showed similar saturation (5 patients / 5,000 spots: median PCC = 0.38 vs. 11-patient model: 0.39), and Martinez *et al.* saturated after training on 10 patients / 10,000 spots (0.33 vs. 0.34). This consistent early plateau indicates that Path2Space achieves stable generalization with a modest amount of training data and that the full model trained on 22 Bassiouni *et al.* patients (56,567 spots) provides sufficient capacity for downstream analyses (Figure S3B).

We next tested whether increasing training diversity across cohorts improves generalization. This was done by training models on Bassiouni *et al.* (our baseline) alone or in combination with HTAN and/or Martinez, validating all models on the HEST dataset. Multi-cohort training showed very modest improvements (Bassiouni alone: median PCC = 0.42; +HTAN: 0.45; +Martinez: 0.44), with performance saturating when all three cohorts were combined (0.44) (Figure S3C).

To specifically examine cross-subtype generalization, we compared the TNBC-only trained model (Bassiouni) versus the mixed-subtype trained model (Bassiouni+HTAN), validating on non-TNBC HEST and TNBC Martinez cohorts. Results revealed a trade-off between subtype specificity and generalization: mixed-subtype training improved performance on non-TNBC HEST samples (TNBC-only median PCC : 0.42 vs. mixed-subtype: 0.46, $p < 0.0001$) but modestly reduced accuracy on TNBC-specific Martinez samples, as expected (0.36 vs. 0.32, $p < 0.0001$; Figure S3D).

Finally, applying the breast-trained model to 13 cancer types in the HEST cohort revealed strong tissue specificity: performance was markedly higher in breast than in any other cancer type. This indicates that while the model generalizes well across breast cancer subtypes, it does not transfer across cancer types, and predicting expression in new cancer types will require cancer-specific model retraining (Figure S3E).

Data augmentation

To evaluate whether augmentation could benefit Path2Space, we generated five augmented variants per training patch (color jittering, rotations, stain transfer) and pre-extracted CTransPath embeddings for each, expanding the training set from 56,567 to 282,835 patches. Augmentation showed modest improvement in cross-validation (median PCC : 0.37 $\rightarrow$ 0.38, $p < 0.05$) but no significant benefit on external validation (median PCC : 0.35 for both datasets, $p = 0.42$; Figure S3F). Augmentations were implemented using torchvision.transforms (version 0.15.2) with ColorJitter (brightness=0.2, contrast=0.2, saturation=0.2, hue=0.1) and RandomRotation (degrees=15). Stain transfer was performed using the StainTools Python package (version 2.1.2).

Model evaluation

Performance metrics

We evaluated gene-wise prediction accuracy by first calculating the Pearson correlation between spatially smoothed actual and predicted expression values across all spots within each slide. We then averaged these correlations across genes and slides from the same patient. Finally, for each gene, the correlation values were averaged across all patients. PCCs were computed using SciPy.⁵⁵

We also assessed our ability to classify high or low gene expression. We binarized the gene expression labels according to the median value per slide and then calculated the AUCs for the prediction of the binarized gene expression for each gene, using the predicted gene expression values as classification scores. Similarly to PCC calculation, for a given gene, we first averaged AUCs across slides from the same patient and then averaged across all patients.

Finally, to identify genes whose prediction performance was substantially lower than expected given their expression level, gene-level predictability was quantified as the PCC between predicted and observed expression values across all spots. The relationship between mean gene expression (log1p-transformed total counts) and predictability was modeled using a four-parameter sigmoid function fit with a robust loss function (soft_l1) to reduce the influence of outliers on the saturation curve. Residuals from this model

were standardized to z-scores, and one-sided *p* values were computed assuming a normal distribution. Multiple testing correction was performed using the Benjamini–Hochberg false discovery rate (FDR) procedure. Genes with FDR < 0.1 and z-scores < −2 were classified as unexpectedly unpredictable (Figure S3G).

Quality control criteria

We trained our model using cross-validation on 43 ST samples from 22 patients in the Bassiouni cohort. Model performance for each sample, quantified by the mean PCC between measured and predicted gene expression across all genes, varied considerably. To ensure robust evaluation and address potential sources of variability, we established several quality control (QC) criteria informed by this training set, focusing on histological image quality and gene expression data reliability.

Model performance was strongly associated with image quality metrics, showing negative correlations with hematoxylin intensity (PCC = −0.51) and eosin intensity (PCC = −0.73), and a positive correlation with sharpness (PCC = 0.46). Because Path2Space is intended for clinical diagnostic applications, we benchmarked these QC metrics against diagnostic H&E images.

First, we assessed stain intensity. ST H&E images—optimized primarily for sequencing alignment rather than diagnostic histopathology—exhibited substantially higher hematoxylin and eosin intensities than clinical H&E slides. To align training data quality with clinical standards, we excluded any slide whose eosin or hematoxylin intensity exceeded the mean of the reference distribution by more than two standard deviations, with the reference distribution estimated from breast cancer diagnostic H&E images in TCGA (yielding exclusion thresholds of 0.023 and 0.053 for eosin and hematoxylin, respectively; Figure S2A).

Second, for image sharpness, ST images were substantially less sharp than diagnostic slides. Sharpness values in the training set exhibited a bimodal-like distribution (Figure S2B), with the lower mode corresponding to samples with poorer model performance. A sharpness threshold of 6×10^{-4} was therefore applied to exclude low-quality samples.

Third, for gene expression quality, we applied established metrics including mean total UMI counts per spot and mean mitochondrial and hemoglobin gene counts per spot. Filtering thresholds were defined from the extreme 1% tails of each metric's distribution within the Bassiouni cohort. Samples with mean total counts per spot below the 1st percentile (< 6,194 reads) or with mean mitochondrial or hemoglobin gene counts above the 99th percentile (> 8.4 and $> 0.7 \log_{10}$ reads per spot, respectively) were excluded.

Taken together, the application of these three QC criteria retained 22 of 43 samples in the Bassiouni cohort. In the independent ST validation cohorts, all five HEST samples passed QC, identifying it as the highest-quality cohort with the best model performance. Four of five samples in the Martinez *et al.* cohort passed QC, whereas nine of 44 samples in the HTAN cohort were retained, indicating comparatively lower data quality in this cohort. Importantly, Path2Space and all 16 other benchmarked methods performed significantly better (Mann–Whitney U test, $p < 4 \times 10^{-13}$ for all methods) on QC-retained samples compared to the unfiltered cohorts (Figure S2C). This finding also held true when evaluating Path2Space with six different foundation model feature extractors ($p < 2 \times 10^{-6}$ for all; Figure S2D), confirming that the established criteria reliably captured factors affecting prediction quality across diverse analytical approaches and model architectures. When applied to the TCGA cohort for SpatioType analysis, QC filtering retained 784 of 853 patients. SpatioType assignments showed 99% concordance with the original labels, and the *Immune-Inactive* SpatioType remained the poorest-prognosis group ($p = 0.006$, log-rank; $p = 0.002$ vs. *Immune-Modulated*).

Histological image quality was assessed on color-normalized tiles. Image sharpness was quantified by converting tiles to grayscale and computing the variance of the Laplacian operator as a focus metric. Stain intensities were measured by transforming images from RGB to hematoxylin-eosin-DAB (HED) color space using optical density deconvolution. Mean hematoxylin and eosin channel intensities were calculated within tissue regions identified by Otsu thresholding applied to grayscale images. Image processing was performed using scikit-image (version 0.25.2).

Gene expression quality control metrics—including total reads per spot and the proportion of counts from mitochondrial and hemoglobin genes—were computed using the `sc.pp.calculate_qc_metrics` function in Scanpy (version 1.11.4).

Tissue preservation analysis

To further investigate the unexpected similarity in performance between FF and FFPE tissues despite Path2Space being trained exclusively on FF slides, we examined differences in nuclear morphology and their associations with prediction accuracy.

To evaluate nuclear morphology, we performed segmentation using CellPose⁵⁶ and extracted eight quantitative features: nuclear area, equivalent diameter, circularity, solidity, eccentricity, aspect ratio, nearest neighbor distance, and local nuclear density (computed within 50 μm radius). Sample-level feature distributions were summarized using mean values. FF and FFPE slides showed characteristic differences in nuclear morphology, with FFPE slides exhibiting smaller and more circular nuclei (lower eccentricity).

To assess whether morphology is associated with model performance, we fit linear mixed-effects models using the MixedLM function from the statsmodels package in Python. The dependent variable was sample-level prediction accuracy (mean Pearson correlation across genes). We tested each morphological feature in separate models with fixed effects including the feature (standardized) and preservation method; dataset/cohort was included as a random intercept to account for batch effects. Models were fit using restricted maximum likelihood (REML). Multiple testing correction across the eight morphological features was performed using the Benjamini–Hochberg false discovery rate (FDR) procedure.

Among morphological features, nuclear eccentricity was the only feature significantly associated with prediction accuracy when controlling for preservation method and correcting for multiple testing ($\beta = -1.57$, FDR $q = 6.5 \times 10^{-7}$), with samples exhibiting lower eccentricity (more circular nuclei) achieving better performance. However, the range of morphological features observed in FFPE samples fell almost entirely within the range present in the FF training dataset. This indicates that the model encountered similar morphologies during training, explaining the consistent generalization across preservation methods.

Confounder analysis

To determine whether Path2Space predictions reflect genuine biological variation rather than technical or compositional artifacts (Table S2), we computed gene-level partial correlations across all cohorts controlling for three potential confounders: hematoxylin stain intensity, cancer cell fraction, and total RNA content per spot.

Hematoxylin intensity, which is directly observable in the input image, correlates with both measured and predicted expression ($r = 0.22$ and 0.30). However, after controlling for this factor, 99.7% of genes (14,021/14,068) retained significant positive partial correlations ($FDR < 0.05$; median partial $r = 0.30$), indicating that predictions capture variation independent of global staining intensity.

After controlling for cancer cell fraction, despite correlations with both measured and predicted expression ($r = 0.52$ and 0.47), 98.8% of genes (13,897/14,068) retained significant partial correlations ($FDR < 0.05$; median partial $r = 0.11$), demonstrating that predictions capture spatial expression patterns beyond tumor density.

Finally, when adjusting for total RNA content, 98.0% of genes (13,788/14,068) remained significant ($FDR < 0.05$; median partial $r = 0.13$ in cross-validation and 0.14 in external validation), despite strong correlations of both observed and predicted expression with RNA abundance ($r = 0.54$ and 0.44). This confirms that Path2Space captures gene-specific spatial variation independent of global RNA yield.

Together, these analyses establish that Path2Space learns biologically meaningful spatial expression variation rather than technical or compositional artifacts.

Hematoxylin intensity per spot was computed as described in the image quality control section. Cancer cell fraction per spot was estimated by applying SpaCET to the measured spatial transcriptomics data. Total RNA content was defined as $\log_{10}(1 + \text{total counts})$ per spot. Partial correlations were computed per gene within each slide using residual-based regression. Per-slide correlations were aggregated using spot count-weighted means, and p values were combined using Stouffer's method with weights proportional to $\sqrt{n_{\text{spots}}}$. FDR correction was applied across genes using the Benjamini–Hochberg procedure.

Cell type-stratified analysis

To evaluate whether Path2Space predictions capture gene-specific variation within distinct cell types, we employed three complementary analytical approaches:

Spatial neighborhood analysis

For each slide, we constructed a k -nearest neighbor spatial graph ($k=6$) based on spot coordinates using Squidpy,⁵⁷ then applied Leiden clustering (resolution = 0.3) to identify spatially coherent neighborhoods. Each neighborhood was assigned a dominant cell type based on mean deconvolution estimates across constituent spots. We computed PCC between predicted and observed gene expression within each neighborhood, including only neighborhoods with ≥ 10 spots to ensure sufficient statistical power.

Pathologist-annotated cell types

We evaluated prediction accuracy using pathologist-assigned cell-type labels for individual spots, computing PCC for genes within each cell-type group.

All four ST validation cohorts were used for transcriptionally weighted and spatial neighborhood analyses, with correlations hierarchically averaged as described for whole-tissue analyses: computing mean correlations per gene within patients, then across patients within cohorts, and finally across cohorts. Pathologist-defined analyses used spot-level cell-type annotations from HEST sample TENX39, computing PCC within each cell-type group.

Benchmarking

Highly variable gene selection

Highly variable genes (HVGs) were identified following the protocol established in the published benchmarking study.³⁰ For each tissue section, the top 1,000 HVGs were computed using *scran*'s *modelGeneVar* and *getTopHVGs* functions. HVGs from all sections were then combined (union), and genes expressed in fewer than 70% of spots across all sections were removed. This protocol yielded 785 HVGs for the legacy ST dataset as reported in the original benchmark, and 988 HVGs when applied to the Bassiouni *et al.* Visium cohort.

Benchmarking on Visium data

We benchmarked Path2Space against 16 state-of-the-art spatial transcriptomics prediction models using the Bassiouni *et al.* Visium cohort: ST-Net,¹⁰ BLEEP,¹¹ STco,⁵⁸ mclSTExp,⁵⁹ EGNv2,³¹ EGGN,³¹ HisToGene,⁶⁰ HistoSPACE,⁶¹ TRIPLEX,⁶² STFlow,⁶³ SEPAL,⁶⁴ DeepSpaCE,¹⁴ DeepPT,⁸ STimage,¹² Hist2ST,¹³ and a Vision Transformer (ViT) baseline (denoted LocalNet). All models were reimplemented and trained under a standardized configuration. All models were trained using leave-one-patient-out cross-validation, and external validation was performed on the HEST, Martinez *et al.*, and HTAN cohorts. Performance was quantified using PCCs between predicted and raw, unsmoothed measured expression values for all methods, including Path2Space. Path2Space predictions were evaluated both with and without spatial smoothing post-processing (smoothed predictions vs. unsmoothed labels, and unsmoothed predictions vs. unsmoothed labels). To assess robustness to normalization strategy, all models were additionally trained on log-transformed CPM-normalized counts. For genome-wide prediction ($n = 14,068$ genes), the top four performers from the HVG benchmark (mclSTExp, STFlow, BLEEP, and EGNv2, identified based on mean PCC averaged across cross-validation and external validation) were retrained and compared. Statistical comparisons between methods used two-sided Mann–Whitney U tests.

Training protocol for Visium benchmark

All models were implemented in PyTorch 2.4.1 and trained on a single NVIDIA RTX A5000 GPU with learning rate = 1×10^{-4} and max_epochs = 200. Early stopping (patience = 20) monitored Pearson correlation for regression models or loss for contrastive/generative models. Learning-rate scheduling used ReduceLROnPlateau (patience = 5, factor = 0.1) for regression models or CosineAnnealingLR ($\eta_{\min} = 10^{-6}$) for contrastive/generative models. Slide-based models used batch size = 1; patch-based models used batch size = 128. Gene expression values were log-transformed ($\log_{10}(\text{count} + 1)$) without library size normalization. For multi-modal contrastive models (BLEEP, STco, mclSTExp), temperature was set to 1.0.

Feature extraction for Visium benchmark

Feature extraction for methods using pathology foundation models was performed using the Trident library (<https://github.com/mahmoodlab/TRIDENT>). All other implementation details were reproduced following the original publications.

Benchmarking on legacy ST data

To compare against published benchmarks, we followed the evaluation protocol of a recent comprehensive benchmarking study,³⁰ which compared 11 spatial transcriptomics predictors on a legacy (pre-Visium) breast cancer ST dataset.³ Ground truth gene expression labels and predictions from all 11 benchmarked models were obtained directly from the publicly released data (<https://doi.org/10.5281/zenodo.14602489>). Path2Space was trained using the identical 785-gene HVG panel and cross-validation splits. All evaluations—including Path2Space—were performed against raw, unsmoothed measured expression values, consistent with the original benchmarking protocol, and prediction accuracy was quantified by PCC.

Cell type deconvolution

Deconvolution model training

Tile images matching the size of Visium spots (pseudo-spots) were extracted from each ROI, and the fractions of each cell type within each image were computed based on their nuclei annotations. To train and evaluate the predictor, we used Path2Space to generate spatial gene expression profiles for the pseudo-spot images. These gene expression profiles served as input to a MLP regressor tasked with predicting cell type fractions.

Within each training fold, we normalized Path2Space-inferred gene expression data to a range of 0–1 using Scikit-learn's MinMaxScaler. We then selected the top 1,000 genes with the highest variance across tumor classes, applying ANOVA F-statistics with SelectKBest and f_{classif} .

The MLP regressor was designed with three output nodes, predicting the fractions of cancer cells, lymphocytes, and stromal cells. Hyperparameters—including learning rate, backpropagation algorithm, momentum, activation functions, number of layers, iterations, and validation split—were fine-tuned through five-fold cross-validation within the training fold. This optimization process, guided by the regression F-score, was performed using Scikit-learn's GridSearchCV.

We used 5-fold cross-validation, keeping all slides from each patient in the same fold to prevent data leakage and ensure model generalizability. For TCGA cross-validation assessment, we combined predictions from all test folds and computed PCC. For test set predictions, we averaged the outputs from all five models to estimate cell type fractions.

Predicted cell type fractions and labels from each fold were aggregated across all test folds to evaluate performance. For external validation, predictions were averaged from the five cross-validation models and applied to H&E images and cell type annotations from the pathologist-annotated HEST samples (Figure S5A).

Model interpretability analysis

To analyze model performance and identify key predictive genes, we aggregated feature sets (genes) from all five trained models (across each of the five folds). Feature importance for each cell type was computed using Feature Attribution via Weight Propagation—a method of linear model interpretation through layer-wise weight analysis.⁶⁵ For genes appearing in more than one fold, importance scores were averaged across models.

Pathway enrichment

We tested gene enrichment using cell type signatures from the SpaCET³⁴ R package, focusing on breast cancer, lymphocytes, and stromal cells (CAFs and macrophages). From each signature, we selected the top 50 most highly expressed genes. Using GSEAPY,⁶⁶ we performed pre-ranked GSEA by ranking genes based on their activation scores and using SpaCET cell type signatures as gene sets. We then assessed enrichment by testing the activation scores from each cell type against all three signatures.

ST clusters

We divided each TCGA slide into contiguous pseudo-spots, yielding approximately 6.7 million pseudo-spots across the 1,096-slide cohort. To identify biologically meaningful spatial patterns while reducing technical noise inherent to individual pseudo-spot measurements, we employed a two-step aggregation approach:

Within-slide domain detection

In the first step, we applied SpaGCN to each slide individually to group pseudo-spots into spatially coherent domains—contiguous tissue regions within each slide that exhibit spatially and transcriptomically coherent gene expression profiles. This yielded an average of 6.6 domains per slide (range: 4–11), reducing the 6.7 million pseudo-spots to 7,295 spatial domains across the cohort. Gene expression was averaged within each domain to generate domain-level expression profiles.

Cross-patient clustering

In the second step, to ensure consistent cluster labeling across tumor slides from different TCGA patients, we normalized the domain-averaged expression profiles using Seurat's log-normalization method and jointly clustered all domains from all TCGA slides using Seurat's *FindClusters* function, identifying 11 ST clusters that were consistent across the TCGA cohort. Each domain was assigned to one of these 11 cluster labels, resulting in unified transcriptomic cluster labels that were spatially and biologically consistent across all samples.

This aggregation strategy serves two purposes: (1) it reduces technical noise inherent to individual pseudo-spot measurements, and (2) it identifies biologically meaningful spatial units—coherent tissue regions that represent distinct microenvironmental niches within tumors.

Notably, while PERMANOVA analysis detected a statistically significant association between clinical subtype and spatial cluster composition ($p < 0.001$), clinical subtype explained less than 1% of variance in spatial patterns, indicating that spatial architecture provides additional biological insight beyond receptor status, consistent with recent ST studies.^{1,3,67,68}

External cohort transfer

To transfer these spatial clusters to independent datasets (TransNEO and IMPRESS), we followed the same pipeline. First, we applied SpaGCN to infer spatial domains from gene expression data in the TransNEO and IMPRESS cohorts, which contained treatment response data for both chemotherapy and trastuzumab. To align domain expression profiles between each cohort and TCGA, we used Seurat's anchor-based integration method (*TransferData* function), mitigating batch effects and ensuring consistent spatial cluster assignment.

Differential expression

To identify cell type-specific gene expression programs, we performed differential expression analysis using the *FindMarkers* function from the Seurat package, specifying the logistic regression test and applying it to log-normalized transcriptomic profiles. To account for technical variation and batch effects, slide identity was incorporated as a latent variable in the model. This approach enabled robust detection of transcriptional differences associated with distinct cellular phenotypes, while controlling for variability arising from sample preparation and imaging conditions.

Pathway activity scoring

A similar approach was applied for pathway enrichment analysis. Instead of raw gene expression values, we utilized single-sample gene set enrichment analysis (ssGSEA), implemented in GSEAPY, to transform the expression profiles into NES based on the MSigDB Cancer Hallmark gene sets. Pathway scores were averaged per cluster and slide, and fold changes were computed relative to the slide's overall pathway activity using linear regression.

Cell type composition

To profile the cellular composition of ST clusters, we aggregated predicted cell type abundances across all slides assigned to each cluster. For each cluster, we computed the mean proportional abundance of each cell type, yielding a representative cellular profile. These profiles were then used to define a compositional signature for each cluster type. As our model achieves the highest accuracy for cancer, lymphocytic, and stromal cell populations—the predominant constituents of the tumor microenvironment—we restricted our analysis to these three cell types. Reported values represent their relative proportions, normalized such that their combined abundance sums to one, thereby facilitating direct comparisons across clusters and experimental conditions.

Ligand-receptor analysis

We applied a ligand-receptor interaction analysis inspired by the differential analysis pipeline from SpatialDM,⁶⁹ utilizing a z-score-based method. Global z-scores of bivariate Moran's I statistics were computed for each slide using SpatialDM, quantifying the spatial co-expression of ligand-receptor pairs across tumor regions.

For each ligand-receptor interaction, a linear regression model was fit, with cluster proportions serving as the independent variables (features) and the interaction z-scores as the response. Significant ligand-receptor pairs were identified through chi-square goodness-of-fit tests, applying a false discovery rate (FDR) threshold of < 0.05 . The regression coefficients were further used to estimate the extent of ligand-receptor co-expression within individual clusters, offering insights into spatial interaction dynamics throughout the tumor microenvironment.

SpatioType analysis

Hierarchical clustering

We performed hierarchical clustering on the patient-cluster proportion matrix to identify distinct *SpatioTypes*. First, we standardized the input matrix using z-score scaling for each feature (cluster) to ensure equal contribution to the clustering process.

We computed a distance matrix using Euclidean distances between the scaled feature vectors. Agglomerative hierarchical clustering was then performed using Ward's minimum variance method (ward.D2), which minimizes the total within-cluster variance at each merging step to produce compact, spherical clusters.

Cluster optimization

We determined the optimal number of clusters (k) by applying multiple statistical criteria. The within-cluster sum of squares (WSS) was computed for $k = 1-10$ using the *factoextra* package (v1.0.7), and the elbow method was used to identify diminishing WSS reduction. The Davies-Bouldin Index (DBI) was calculated with the *cluster* package (v2.1.4), where minimum DBI indicates optimal separation. Silhouette analysis was performed to assess cluster cohesion and separation. Additionally, the Proportion of Ambiguous

Clustering (PAC) was computed with ConsensusClusterPlus (v1.62.0), defined as the proportion of sample pairs with consensus values between 0.1 and 0.9. A PAC value of 0.128 indicated strong cluster stability at $k = 5$, which was selected as the final clustering solution for the TCGA BRCA cohort.

Survival analysis

We analyzed overall and disease-free survival data from TCGA.⁷⁰ Survival analyses were performed using the survival package (version 3.5.7) in R, implementing log-rank tests and Cox proportional hazards models.

For clinical subtype annotations, we used receptor status data from TCGA.⁷¹ We classified patients as HER2+ when HER2 status was marked 'positive'. Among HER2– patients, those with positive status for either ER or PR were classified as HR+. Patients negative for all three receptors (HER2, ER, and PR) were classified as triple-negative breast cancer (TNBC). We excluded patients with missing receptor status data or those marked as 'equivocal' in TCGA annotations.

External cohort projection

To assign *SpatioTypes* to METABRIC patients, we applied Path2Space to infer ST cluster compositions from H&E images. We computed the centroid of each *SpatioType* in TCGA and assigned each METABRIC patient to the *SpatioType* with minimum Aitchison distance on cluster proportions, using the compositions R package (version 2.0-6). Assignment confidence was quantified as one minus the normalized Shannon entropy of the softmax-transformed distances to centroids. Patients with low-confidence assignments (confidence score < 0.5, indicating assignment entropy exceeding 50% of maximum uncertainty; $n = 17$, 11%) were excluded.

SPAND analysis

SPAND calculation

SPAND was defined as $\frac{-\text{Moran's } I(\text{gene})}{\text{mean}(\text{gene})}$, where the Moran's I and the mean were calculated on the predicted expression of a given gene across pseudo-spots. Moran's I , which quantifies spatial autocorrelation, was computed using the PySAL⁷² (24.01) package (esda module) with spatial weights derived from the pseudo-spot grid. The weights matrix was generated with the lat2W function, applying a rook's case contiguity structure based on the grid dimensions (number of rows and columns), connecting each pseudo-spot to its four cardinal neighbors (top, bottom, left, right). To ensure robustness and comparability across spots with varying numbers of neighbors, the weights matrix was row-standardized, ensuring equal contributions to the final Moran's I calculation.

HER2 SPAND. To estimate HER2 expression in cancer cells, we applied Path2Space-inferred spatial gene expression to compute the normalized enrichment score (NES) for the 'ErbB Signaling Pathway (WP673)' from the WikiPathways library.⁷³ NES values were calculated per pseudo-spot using ssGSEA implemented in GSEAPY. To isolate HER2 activity specifically in cancer cells, NES values were adjusted by dividing by the cancer cell fraction, estimated by the cell type abundance model at each pseudo-spot. This yielded HER2 expression per cancer cell. SPAND analysis was performed on these normalized values.

Treatment response prediction

Cluster-based model

Cluster proportions were used as input features to predict response. A pipeline was trained consisting of feature standardization using Scikit-learn's StandardScaler, followed by L1-regularized logistic regression. During cross-validation, StandardScaler was fitted to each training fold and applied to both the training and test folds to ensure consistent scaling. Model scores and labels were aggregated across all test folds to evaluate performance.

For external validation, predictions from the five cross-validation models were averaged to obtain the final prediction score. In cases where clusters from the TransNEO development cohort were absent in the IMPRESS cohort, their values were set to zero to maintain input consistency.

Combined trastuzumab model

The combined model predicted trastuzumab response by averaging the prediction scores from the Cluster Model and HER2 SPAND. SPAND scores were standardized using StandardScaler in the same cross-validation pipeline to ensure consistency. For the IMPRESS cohort, scaling factors (mean and standard deviation) were averaged across the five cross-validation folds. The final prediction score was the average of the cluster model and scaled SPAND score.

Control models

To evaluate the predictive power of a model based on inferred bulk gene expression, we developed a control model with a similar architecture to Path2Space. This bulk expression prediction model was trained on the BRCA-TCGA cohort ($n = 869$ slides with matched H&E images and bulk gene expression data) to predict bulk gene expression values from H&E slides.

The model's architecture mirrors that of Path2Space, utilizing the same image preprocessing and feature extraction pipeline. An MLP regressor was trained to predict bulk gene expression for each tile. Predictions were then averaged across all tiles from the same slide to generate inferred bulk expression values for each sample.

Training was conducted using a 5x5 nested cross-validation strategy. The resulting model was applied to the TransNEO and IMPRESS cohorts to infer bulk expression values. For the TransNEO cohort, a logistic regression model was trained to predict response based on the inferred bulk values. The data were first scaled using Scikit-learn's StandardScaler and the top 11 features

were selected via SelectKBest. An L1 penalty was applied to the logistic regression model to enhance feature sparsity. This model was trained in five-fold cross-validation on both trastuzumab-treated and chemotherapy datasets and validated on the IMPRESS cohort.

To assess the feasibility of predicting treatment response directly from slide images, we developed a second control model using the same image preprocessing and feature extraction pipeline as Path2Space. This direct prediction model bypasses gene expression inference by training an MLP classifier to predict response directly from extracted image features.

For model training, each slide was partitioned into tiles of 500x500 pixels at 20x magnification, with each tile assigned the label of the entire slide. The model was trained using a 5x5 nested cross-validation strategy on the TransNEO cohort and validated on the IMPRESS cohort to evaluate its generalizability to independent datasets.

Comparison methods

We compared our models to previously published models which also predicted trastuzumab and chemotherapy response in TransNEO and the IMPRESS cohorts. We obtained the scores for the Sammut-ML model²⁸ from transneo-diagnosis-MLscores.tsv downloaded from <https://github.com/cclab-brca/neoadjuvant-therapy-response-predictor>. We used the scores for the Clinical+D-NA+RNA+DigPath model. We collected the scores for the ENLIGHT model⁴⁷ from <https://ems.pangeabiomed.com/>. For the Huang *et al.* model,²⁹ we extracted the AUCs directly from Table 2 in the paper, as the scores were not provided. We did not report the 95% confidence intervals as these were also not available. We used the AUCs for the model based on the H&E only to ensure a fair comparison with our model. Finally, for the models from Aswolinskiy *et al.*,⁷⁴ we obtained the AUCs and 95% confidence intervals directly from Table 4 in the paper. We compared our method to the models based on the ratio of lymphocytes to tumor in the slide (LTR), the ratio of (inflamed) tumor close to lymphocytes to the overall tumor amount (ITR), the computational tumor infiltrating lymphocytes score (cTILs) and the mitotic rate (MTR) as the number of detected mitoses within tumor regions divided by the tumor area.

Evaluation metrics

For model performance evaluation, we used AUC, AUPRC, as implemented in Scikit-learn, and odds ratio metrics.

To calculate the odds ratio for the Path2Space, bulk and direct model, we first use the Youden's J,⁷⁵ statistic to learn optimal thresholds for each of the methods on the training data (TransNEO). For each method, we identified the threshold that maximized J, defined as:

$$J = TPR - FPR,$$

where TPR is the true positive rate and FPR is the false positive rate.

For the test cohorts (PBCP, IMPRESS and Cedars-Sinai combined), we applied the Haldane-correction^{76,77} to the contingency table to address potentially undefined odds ratio values. This correction involved adding 0.5 to each cell of the contingency table. The corrected odds ratio (OR) was then calculated as:

$$OR = \frac{(TP + 0.5) \times (TN + 0.5)}{(FP + 0.5) \times (FN + 0.5)},$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

We estimated *p* values for each odds ratio using the following steps. Calculate standard error (SE)

$$SE = \sqrt{\frac{1}{TP + 0.5} + \frac{1}{TN + 0.5} + \frac{1}{FP + 0.5} + \frac{1}{FN + 0.5}}$$

Compute test statistic Z: $Z = \frac{\ln(OR)}{SE}$

Determine *p* value: $p = 2 \cdot (1 - \Phi(|Z|))$

Where $\Phi(|Z|)$ is the cumulative distribution function (CDF) of the standard normal distribution.

Computational resources

Analyses were conducted using Python 3.10.8 and R 4.3.2. Data processing was performed with NumPy (1.24.4), Pandas (2.2.2), and scikit-learn (1.5.1). Spatial transcriptomics analysis used Scanpy (1.10.4) and SpaGCN for spatial clustering. Cell type deconvolution was performed with SpaCET. Deep learning models were implemented in PyTorch (2.4.1) with timm (1.0.15) for vision transformer architectures. Pretrained histopathology foundation models (CTransPath, GigaPath, UNI2-h) were used for feature extraction. Image preprocessing, including tile partitioning and stain normalization, was performed using OpenSlide (1.3.1), OpenCV (4.6.0), and Pillow (9.2.0). Survival analyses were performed with lifelines (0.28.0) in Python. Visualizations were generated using Matplotlib (3.7.2), Seaborn (0.13.2), and ggplot2 (3.5.1) in R. Computations were performed on a Linux system equipped with an x86_64 processor (28 physical cores), 251.5 GB RAM, and NVIDIA K80, A100 and V100 GPUs.

QUANTIFICATION AND STATISTICAL ANALYSIS

For spatial gene expression prediction, gene-wise Pearson correlation coefficients (PCC) were calculated between predicted and measured expression values for each slide, then averaged within each held-out patient, and finally across all patients. External validation was performed on independent held-out cohorts (HEST, Martinez *et al.*, HTAN) using quality control thresholds fixed a priori based on the training cohort.

For survival analysis, we used Kaplan–Meier curves with the log-rank test to compare high and low-risk groups. Cox proportional hazards regression was used to assess associations between *SpatioTypes* and disease-free survival, adjusting for clinical covariates. The DeLong test was used to compare AUCs between models. For treatment response prediction, logistic regression models were trained using cross-validation on TransNEO and validated on the independent IMPRESS cohort.

Mann–Whitney U tests were used to compare distributions between groups. Multiple testing correction was performed using the Benjamini–Hochberg procedure. Pearson’s correlation coefficient was used to assess correlations between continuous variables. Differences were considered statistically significant when $p < 0.05$ ($*p \leq 0.05$, $**p \leq 0.01$, $***p \leq 0.001$, $****p \leq 0.0001$). Statistical details of experiments, including exact values of n , can be found in the figure legends and [Results](#) section.

Supplemental figures

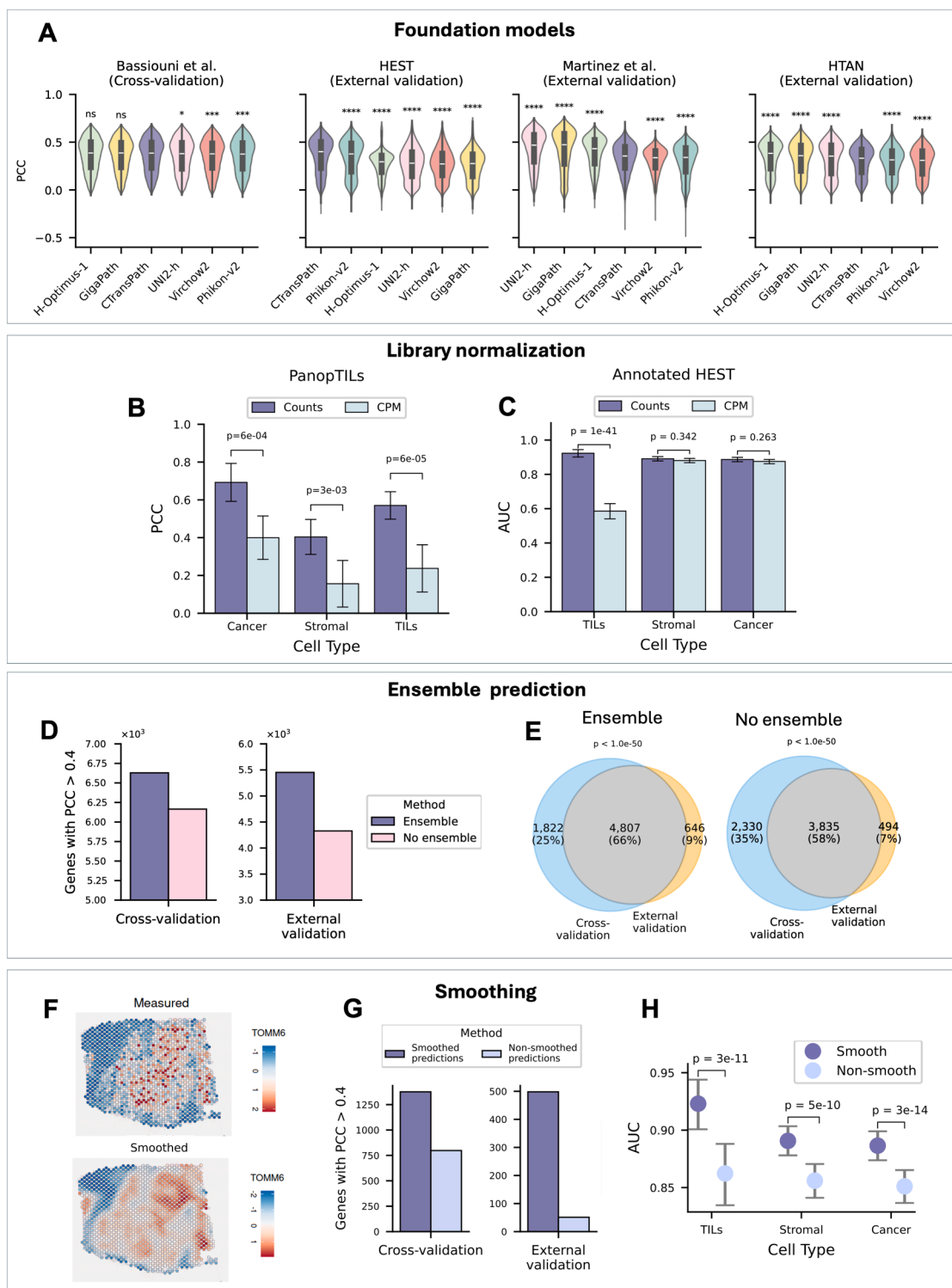


Figure S1. Evaluation of Path2Space methodological components, related to Figure 2

(A) Foundation model performance comparison. Violin plots show gene-level PCC between predicted and measured expression for Path2Space trained with six foundation models (H-Optimus-1, GigaPath, CTransPath, UNI2-h, Virchow2, Phikon-v2) across cross-validation (Bassiouni et al.) and three external validation cohorts (HEST, Martinez et al., HTAN). CTransPath (green) serves as the reference model. Statistical significance versus CTransPath: ns, not significant; * $p < 0.05$, *** $p < 0.001$, **** $p < 0.0001$ (two-sided Mann-Whitney U-test).

(B) Impact of library normalization on cell-type deconvolution in PanopTILs. SpaCET, a deconvolution method that estimates cell-type abundances using single-cell RNA-seq (scRNA-seq)-derived reference signatures, was applied to spatial gene expression predicted by Path2Space from TCGA H&E slides. Bar plots show PCC between SpaCET-estimated and PanopTILs-annotated cell-type fractions for cancer, stromal, and TILs, comparing Path2Space models trained on log-transformed counts (dark purple) versus CPM-normalized values (light purple). p values from two-sided Mann-Whitney U-tests. Error bars indicate standard error.

(C) Impact of library normalization on cell-type classification in annotated HEST. SpaCET deconvolution was applied to Path2Space-predicted spatial gene expression from H&E images. Bar plots show AUC for classifying the presence/absence of TILs, stromal, and cancer cells based on SpaCET-estimated fractions, comparing Path2Space models trained on log-transformed counts (dark purple) versus CPM-normalized values (light purple). p values from two-sided DeLong tests. Error bars indicate standard error.

(D) Ensemble versus single model comparison. Bar plots show the number of genes achieving PCC > 0.4 for ensemble averaging across all cross-validation models (purple) versus the best single model (pink) in cross-validation (left) and external validation (right).

(E) Cross-validation and external validation overlap. Venn diagrams show genes achieving PCC > 0.4 in both validation schemes for ensemble (left) versus single model (right). Ensemble identifies 25% more robustly predicted genes (4,807 vs. 3,835).

(F) Visualization of spatial smoothing of gene expression. An example slide from the Bassiouni et al. cohort displaying the raw expression of *TOMM6* (left) compared with the smoothed expression (right). Colors represent scaled expression levels, with blue indicating low expression and red indicating high expression.

(G) Spatial smoothing impact. Number of genes achieving PCC > 0.4 for smoothed (purple) versus unsmoothed (light blue) predictions in cross-validation (left) and external validation (right), when compared with the unsmoothed labels.

(H) Impact of spatial smoothing on cell-type classification. SpaCET deconvolution using scRNA-seq-derived reference signatures was applied to Path2Space-predicted spatial gene expression, with (dark blue) or without (light blue) spatial smoothing applied to the predictions. Bar plots show AUC for classifying the presence/absence of each cell type on two HEST samples, comparing SpaCET-estimated cell-type fractions to unsmoothed spot-level pathologist annotations. Points indicate AUC values, and error bars show 95% confidence intervals from 1,000 bootstrap iterations. p values from two-sided DeLong tests.

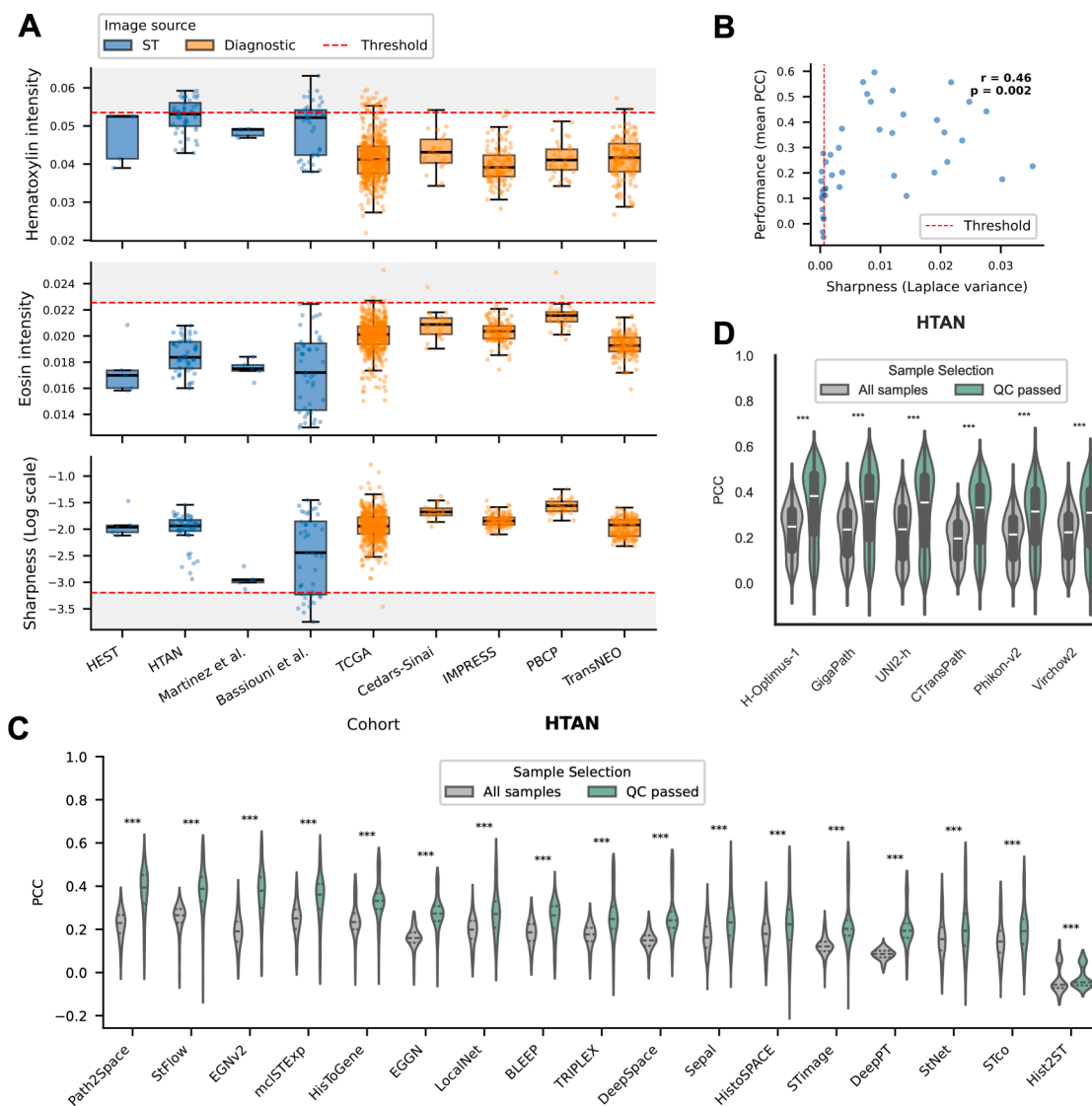
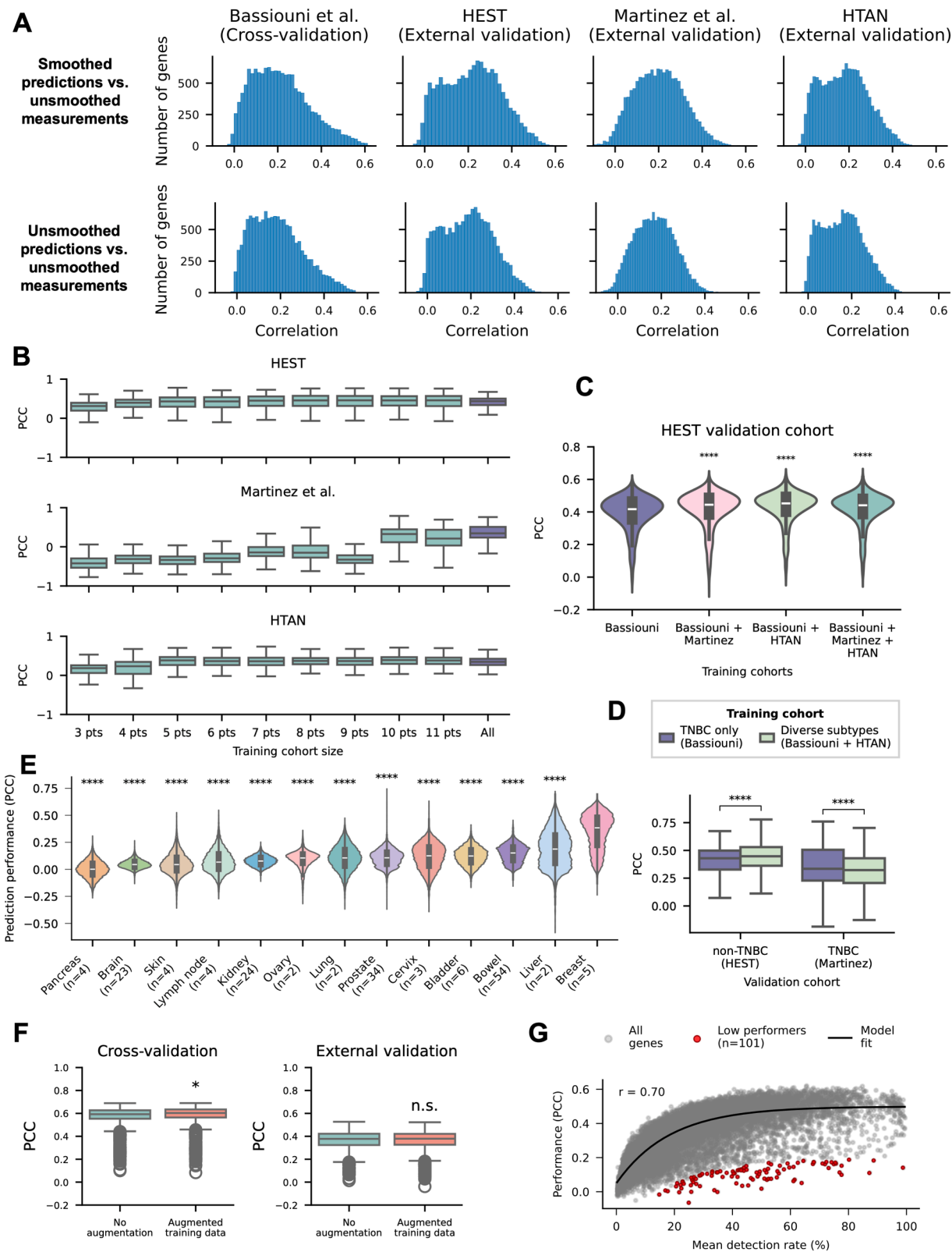


Figure S2. QC analysis, related to Figure 2

(A) Image quality comparison across ST and diagnostic cohorts. Distribution of hematoxylin intensity, eosin intensity, and sharpness ($\log_{10}$ Laplacian variance) across ST cohorts and TCGA diagnostic images. Red dashed lines: QC thresholds. Gray shading: filtered regions. Boxes show quartiles, and individual samples are overlaid.

(B) Image sharpness correlates with model performance. Relationship between image sharpness (Laplacian variance) and prediction accuracy (mean PCC) across training samples. Each point represents one sample from the Bassiouni cohort ($n = 43$). Red dashed line: QC threshold (6×10^{-4}). Pearson correlation coefficient and two-sided p value shown.

(C and D) QC improves performance across all benchmarked methods and feature extractors. Performance comparison on HTAN cohort samples before (gray) and after (green) applying QC filters. (C) Path2Space and 16 other gene expression prediction methods. Violin plots show per-gene PCC distributions. (D) Path2Space with six different foundation model feature extractors (CTransPath is the original implementation). Statistical significance from the Mann-Whitney U-test: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$; ns = not significant. All methods and feature extractors perform significantly better on QC-passed samples.

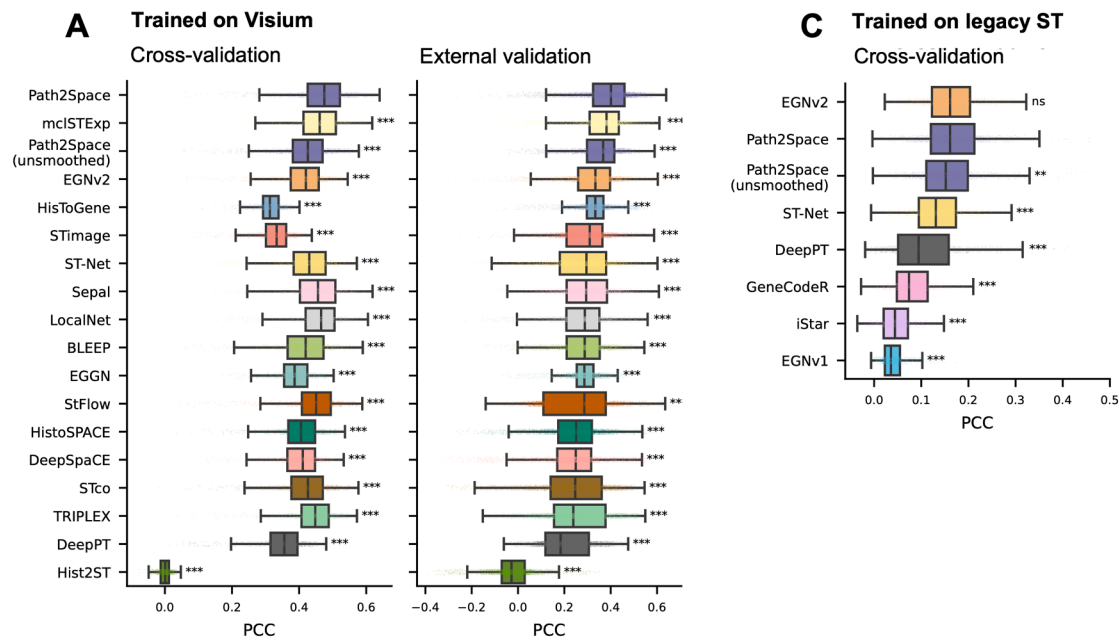


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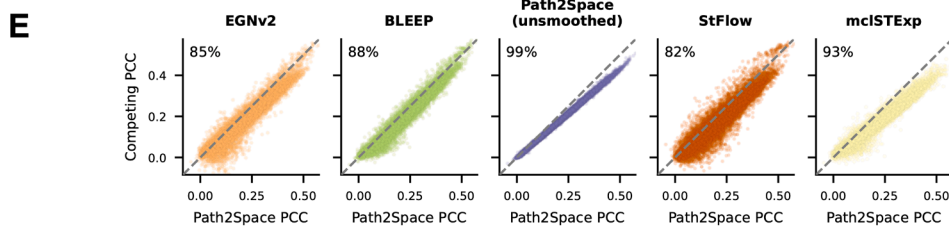
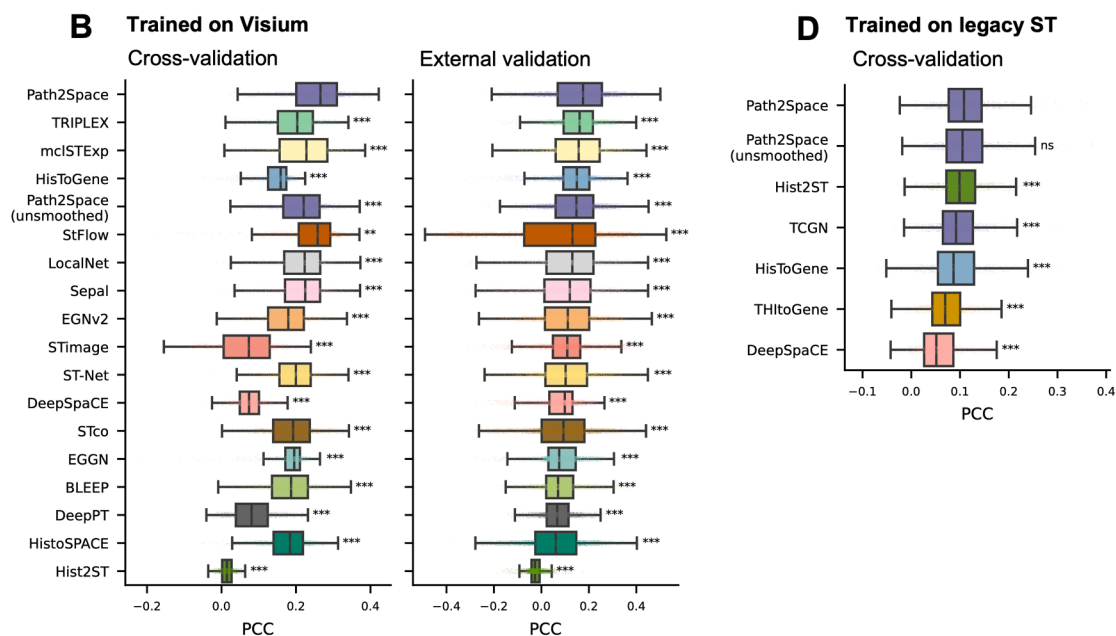
Figure S3. Generalization and robustness of Path2Space predictions across datasets and training settings, related to Figure 2

- (A) Per-Gene PCCs between predicted and measured expression in cross-validation (Bassiouni et al.) and external validation (HEST, Martinez et al., HTAN). Top row: smoothed predictions vs. unsmoothed measurements. Bottom row: unsmoothed predictions vs. unsmoothed measurements.
- (B) Path2Space models trained on subsets of Bassiouni et al. (3–11 patients; 1,000 spots each) were evaluated on three validation cohorts. x axis shows training size, and the purple box shows the full model (43 samples, 22 patients, 56.6k spots).
- (C) Performance on HEST using models trained on Bassiouni alone (TNBC, $n = 43$) or combined with Martinez ($n = 4$) and/or HTAN ($n = 44$). Asterisks show significance versus TNBC-only (**** $p < 0.0001$, Wilcoxon signed-rank test).
- (D) TNBC-only (Bassiouni, $n = 43$, purple) versus mixed-subtype (Bassiouni+HTAN, $n = 87$, green) models evaluated on non-TNBC HEST ($n = 4$) and TNBC Martinez ($n = 4$). **** $p < 0.0001$, Mann-Whitney U-test.
- (E) Breast-trained Path2Space applied to 13 cancer types from HEST ($n = 167$ samples total), ordered by median performance. Asterisks indicate comparison to breast (**** $p < 0.0001$, two-sided Mann-Whitney U-test).
- (F) Comparison of prediction accuracy with and without training data augmentation (5 augmented variants per patch: color jittering, random rotations, and stain transfer, expanding from 56,567 to 282,835 patches). Left: cross-validation shows modest improvement (median PCC: 0.37 vs. 0.38, * $p < 0.05$). Right: external validation shows no significant benefit (median PCC: 0.35 for both, $p = 0.42$). Two-sided Mann-Whitney U-tests.
- (G) Gene predictability correlates with detection rate following a saturation pattern (PCC = 0.70). Red points indicate 101 genes with significantly lower predictability than expected after adjusting for detection rate (FDR < 0.1, $z < -2$). Black line shows a robust sigmoid fit.
- (B)–(F) show results for 988 HVGs. Boxplots indicate medians, interquartile ranges, and $1.5 \times$ IQR whiskers.

Without library size normalization



With library size normalization



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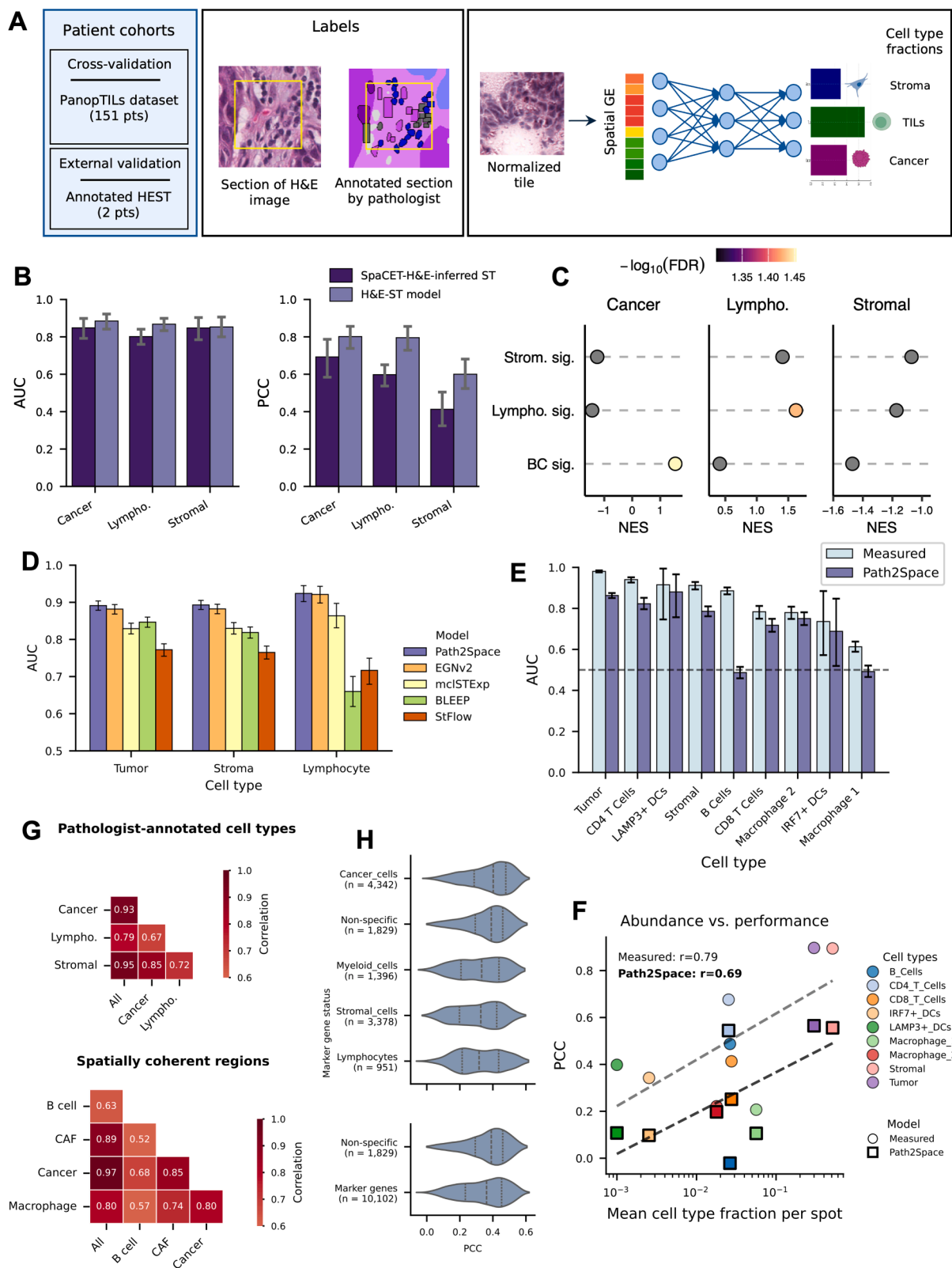
Figure S4. Benchmarking Path2Space against state-of-the-art ST prediction models across multiple datasets and normalization strategies, related to Figure 3

(A and B) Performance comparison of 16 models trained on the Bassiouni et al. Visium breast cancer cohort to predict HVGs ($n = 988$), without library size normalization (A) and with library size normalization (B). Results shown for cross-validation on the Bassiouni et al. cohort (left) and external validation averaged across three independent cohorts: HEST, Martinez, and HTAN (right). Performance measured using PCC between predicted and measured gene expression. Models are sorted by median PCC in the external validation set. Path2Space achieved the highest median PCC in both cross-validation and external validation, with and without spatial smoothing post-processing, across both normalization strategies.

(C and D) Performance comparison on the legacy ST breast cancer dataset under the original preprocessing conditions used by each method. Path2Space was trained on this legacy cohort using the identical set of 785 HVGs and cross-validation protocol from a previous benchmarking study, and its performance is compared against the published results of six other methods evaluated without library size normalization (C) and five other methods evaluated with library size normalization (D). All results are based on cross-validation, with models trained and evaluated using the protocols reported in the original benchmarking study. Models are sorted by median PCC. Path2Space achieved the second-highest median accuracy, with no statistically significant difference from the top-performing method (EGNv2).

All models were trained and evaluated on identical raw, unsmoothed gene expression data to ensure fair comparison. For Path2Space, spatial smoothing was applied only as a post-processing step to predictions, not during training. “Path2Space (unsmoothed)” reports performance without this post-processing, while the standard Path2Space results include spatial smoothing of predictions. Boxplots show the distribution of per-gene PCC values across all predicted genes. Statistical significance was assessed using a two-sided Mann-Whitney U-test comparing each method to Path2Space ($^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; ns, not significant). Boxplots show the median (center line), interquartile range (box), and whiskers extending to $1.5 \times$ IQR.

(E) Pairwise scatterplots comparing per-gene PCC between Path2Space (x axis, one panel per method) and each of the four competing methods from Figure 3B (EGNv2, BLEEP, StFlow, mclSTExp) as well as Path2Space without spatial smoothing (y axis), evaluated across all 14,068 genes and averaged across cross-validation and external validation cohorts. The dashed diagonal indicates equal performance, and points below the line represent genes where Path2Space achieves a higher PCC. The percentage in each panel indicates the fraction of genes for which Path2Space (with smoothing) outperforms the respective competing method.



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Figure S5. Validation of Path2Space cell-type deconvolution across multiple benchmarks and cell-type markers, related to Figure 4

(A) Schematic of the Path2Space model for estimating cell-type fractions. The model was trained using PanopTILs cell-type annotations and externally validated on annotated HEST samples (left). Path2Space estimates per-tile cell-type fractions using a multilayer perceptron (MLP) that takes inferred spatial gene expression from WSIs as input (right).

(B) Performance comparison between two approaches: direct cell-type deconvolution with Path2Space (lighter purple) and SpaCET deconvolution applied to Path2Space-predicted gene expression (dark purple). Analysis was performed on TCGA H&E images with PanopTILs cell-type annotations. Left: classification performance (AUC) for detecting the presence/absence of cell types across individual tissue sections ($n = 20$ tissue sections with ≥ 20 ROIs per section). Right: correlation between actual and predicted cell-type fractions across the same tissue sections. All metrics were computed per tissue section and averaged. Error bars represent standard error across tissue sections.

(C) GSEA showing cell-type signature enrichment scores for each MLP prediction node, from left: cancer, lymphocyte, and stromal cells. (STAR Methods). The x axis shows the normalized enrichment score (NES). The color scale indicates $-\log_{10}$ adjusted p value, with gray denoting non-significant values ($FDR \geq 0.05$).

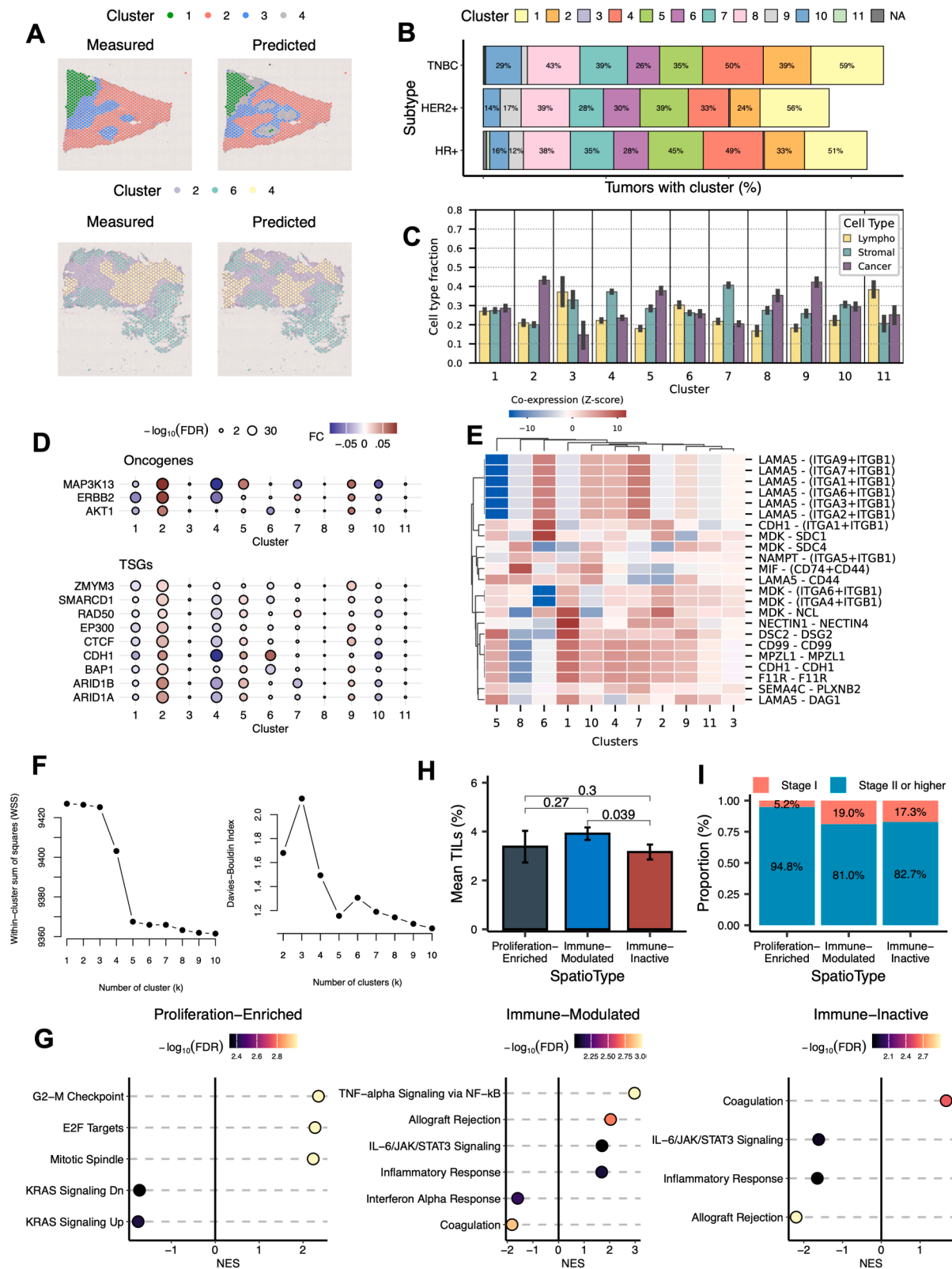
(D) Cellular composition inference from routine H&E images. AUC values for classifying spots in HEST dataset sections with pathologist annotations as containing or lacking specific cell types (tumor, stroma, and lymphocyte) based on SpaCET deconvolution applied to predicted gene expression. The top four competing methods from the benchmarking analysis (Figure 3) were trained to predict the same 14,068 genes and compared against Path2Space: EGNv2 (orange), mclSTExp (yellow), BLEEP (green), and StFlow (red). Path2Space shown in purple. Error bars indicate 95% confidence intervals from 1,000 bootstrap iterations.

(E) AUC values for cell-type prediction accuracy obtained by applying SpaCET deconvolution to measured Visium ST (light blue) or Path2Space-inferred ST (dark blue), evaluated against Xenium-derived ground-truth labels. AUCs were calculated using only spots where a cell type was either absent (0%) or clearly present ($>10\%$). Cell types are ordered by measured ST performance. Error bars denote 95% confidence intervals from 1,000 bootstrap samples. Dashed line indicates AUC = 0.5 (random performance).

(F) Scatterplot showing the relationship between cell-type abundance (x axis, $\log_{10}$ scale) and deconvolution accuracy (y axis, PCC). Circles represent deconvolution based on measured ST, and squares represent Path2Space-inferred ST. Both approaches show abundance-dependent performance (measured: $r = 0.79$; Path2Space: $r = 0.69$). Regression lines depict the association between $\log(\text{abundance})$ and PCC.

(G) Pairwise correlations of gene-wise prediction performance across cell types. For each cell type, PCCs between predicted and measured expression were computed per gene across spots, yielding a performance vector, and heatmap entries report correlations between these vectors for each pair of cell types. Top: pathologist-annotated regions. Bottom: spatially coherent neighborhoods.

(H) Violin plots show per-gene PCC between predicted and measured expression, stratified by marker status from scRNA-seq differential expression. Marker candidates: adjusted $q < 0.05$, $\log_2FC > 0.15$, detection rate difference > 0.01 . Exclusive markers: top $\log_2FC$ exceeds second-highest cell type by ≥ 0.01 . Non-specific genes comprise all remaining genes not meeting marker criteria. Top: exclusive markers by cell type versus non-specific genes. Bottom: all marker candidates versus non-specific genes. Groups are ordered by median PCC, horizontal lines show median and IQR, and sample sizes are indicated on the y axis.



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Figure S6. Characterization of breast cancer spatial transcriptomic clusters, related to Figure 5

(A) Representative outputs from the spatial clustering pipeline applied to the Bassiouni et al. cohort, showing two tissue sections (397D, left; 395D, right). Each tissue section shows a direct comparison between clusters derived from measured ST data (left) and corresponding clusters derived from Path2Space-predicted gene expression profiles (right).

(B) Percentage of tumors containing each ST cluster across clinical subtypes.

(C) Mean fractions of cancer, lymphocyte, and stromal cells across ST clusters, normalized to represent proportions relative to these three major cell populations only.

(D) Dot plot showing the cluster-specific differential expression of breast cancer oncogenes and tumor suppressors across ST clusters. Color indicates average fold change between the cluster and other clusters within slides, and dot size represents significance ($-\log_{10}$ FDR) from two-sided linear regression.

(E) Heatmap of spatially resolved ligand-receptor interactions inferred by SpatialDM (likelihood ratio test, FDR < 0.05, two-sided). Rows show ligand-receptor pairs, columns represent clusters, and colors indicate global Z scores of bivariate Moran's I statistics.

(F) (Left) Elbow plot showing the within-meta-cluster sum of squares (WSS) for different numbers of subgroups (SpatioTypes) (k). The plot indicates that five subgroups provide the optimal balance between complexity and explained variance in the TCGA BRCA cohort. (Right) Davies-Bouldin Index (DBI) calculated across the same range of subgroup numbers. The optimal clustering solution corresponds to the minimum DBI value, confirming that five subgroups yield the best-defined clusters with the highest intra-cluster similarity and inter-cluster separation.

(G) GSEA comparing each SpatioType against all others. Rows show cancer hallmark gene sets, with normalized enrichment scores (NES) on the horizontal axis. Color intensity represents $-\log_{10}$ adjusted p values. Shown are significant results (FDR < 0.05) for proliferative hallmarks in the proliferation-enriched SpatioType and immune hallmarks in the immune-inactive and immune-modulated SpatioTypes.

(H) Mean percentage of TILs (tumor-infiltrating lymphocytes) across SpatioTypes. p values from the one-sided Mann-Whitney U-test.

(I) Percentage of tumors in pathologic Stage I or higher across SpatioTypes.

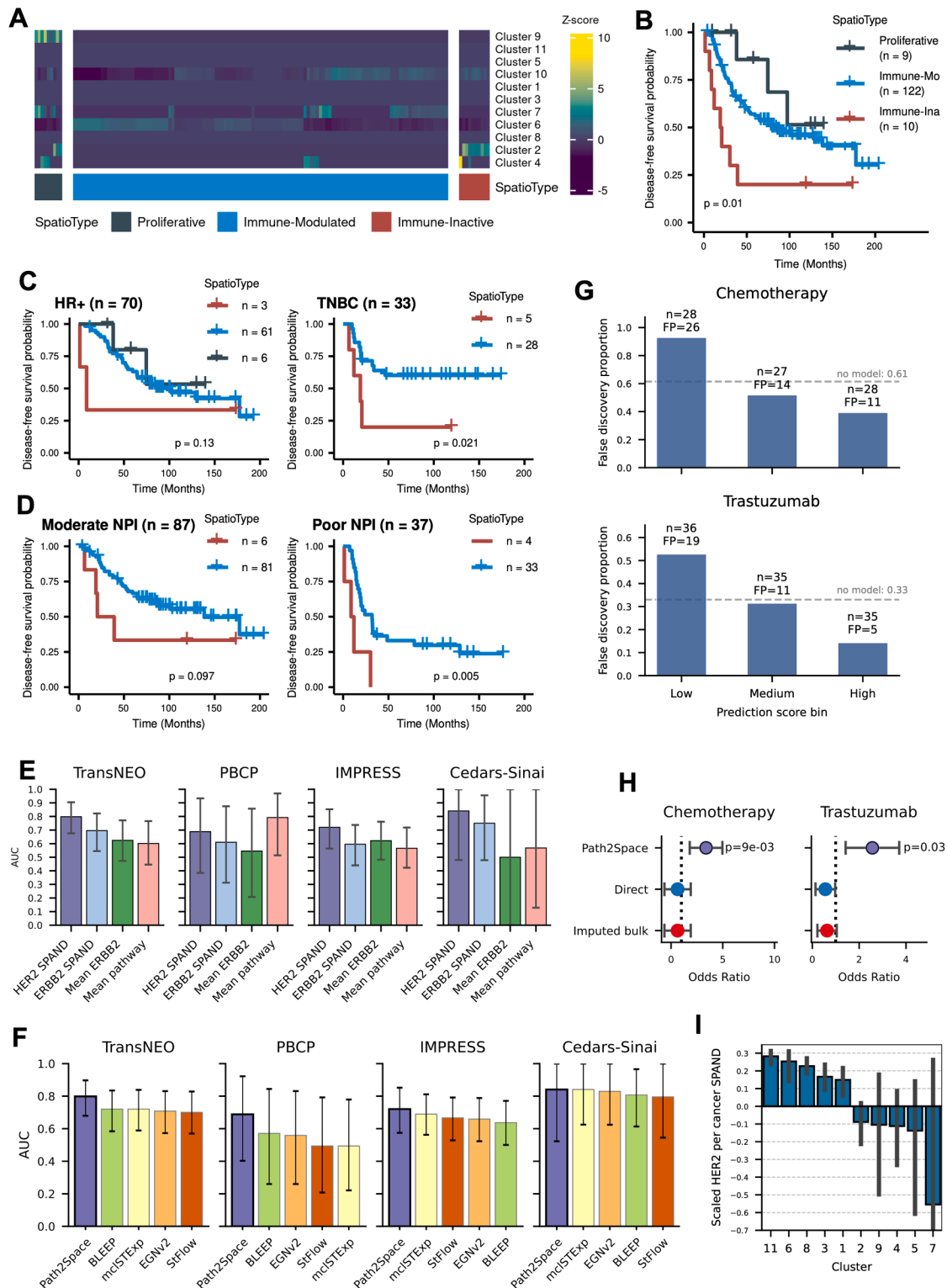


Figure S7. Validation of SpatioTypes and spatial biomarkers for prognosis and trastuzumab response prediction, related to Figures 5, 6, and 7

(A) Heatmap of ST cluster compositions across chemotherapy-treated METABRIC patients ($n = 141$) assigned to TCGA-derived SpatioTypes. Rows represent the 11 ST clusters (Z scored across patients); columns represent individual patients grouped by SpatioType assignment.

(B) Kaplan-Meier analysis of disease-free survival across the three SpatioTypes. p value from two-sided log-rank test.

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- (C) Subgroup survival analysis within HR+ (left; $n = 70$) and TNBC (right; $n = 33$) patients. p values from a one-sided log-rank test.
- (D) Subgroup survival analysis stratified by Nottingham Prognostic Index (NPI): moderate prognosis (NPI 3.4–5.4; left; $n = 87$) and poor prognosis (NPI > 5.4 ; right; $n = 37$). p values from one-sided log-rank tests.
- (E) Predictive performance (AUC) of multiple HER2-related biomarkers for trastuzumab response across four independent cohorts: TransNEO, PBCP, IMPRESS, and Cedars-Sinai. Features include HER2 SPAND (HER2 pathway activity spatial heterogeneity), ERBB2 SPAND (ERBB2 gene expression spatial heterogeneity), mean predicted ERBB2 expression, and mean ERBB2 pathway activity, each normalized by cancer cell fraction. Error bars denote 95% confidence intervals estimated from 1,000 bootstrap iterations.
- (F) Benchmarking HER2 SPAND-based trastuzumab response prediction across spatial gene expression methods. AUC values for predicting pCR to trastuzumab using HER2 SPAND scores derived from gene expression predicted by Path2Space and four top-performing competing methods from our benchmarking analysis (Figure 3): BLEEP, mcIStExp, EGNv2, and StFlow. Results shown across four independent breast cancer cohorts: TransNEO ($n = 61$), PBCP ($n = 18$), IMPRESS ($n = 62$), and Cedars-Sinai ($n = 26$). Path2Space achieves the highest or comparable performance across all cohorts. Error bars represent 95% confidence intervals from 1,000 bootstrap iterations.
- (G) False discovery proportion across prediction score bins. Patients in the external validation cohorts (PBCP, IMPRESS, and Cedars-Sinai combined) were stratified into tertiles based on Path2Space prediction scores for chemotherapy (top) and trastuzumab (bottom) response. Bars show the proportion of non-responders (false discoveries) within each score bin. Dashed line indicates the expected false discovery proportion without a model (overall non-response rate). Patient counts (n) and false positive counts (FP) are annotated per bin.
- (H) Forest plot showing ORs for treatment response in predicted responders versus non-responders, using the decision threshold defined in the STAR Methods. Results are shown for chemotherapy in the PBCP and IMPRESS cohorts (left) and trastuzumab in the PBCP, IMPRESS, and Cedars-Sinai cohorts (right). Error bars denote standard errors, and p values are derived from the Wald test on the log OR.
- (I) Mean HER2 SPAND scores for each of the clusters in trastuzumab-treated patients from all cohorts. Error bars denote the standard error.